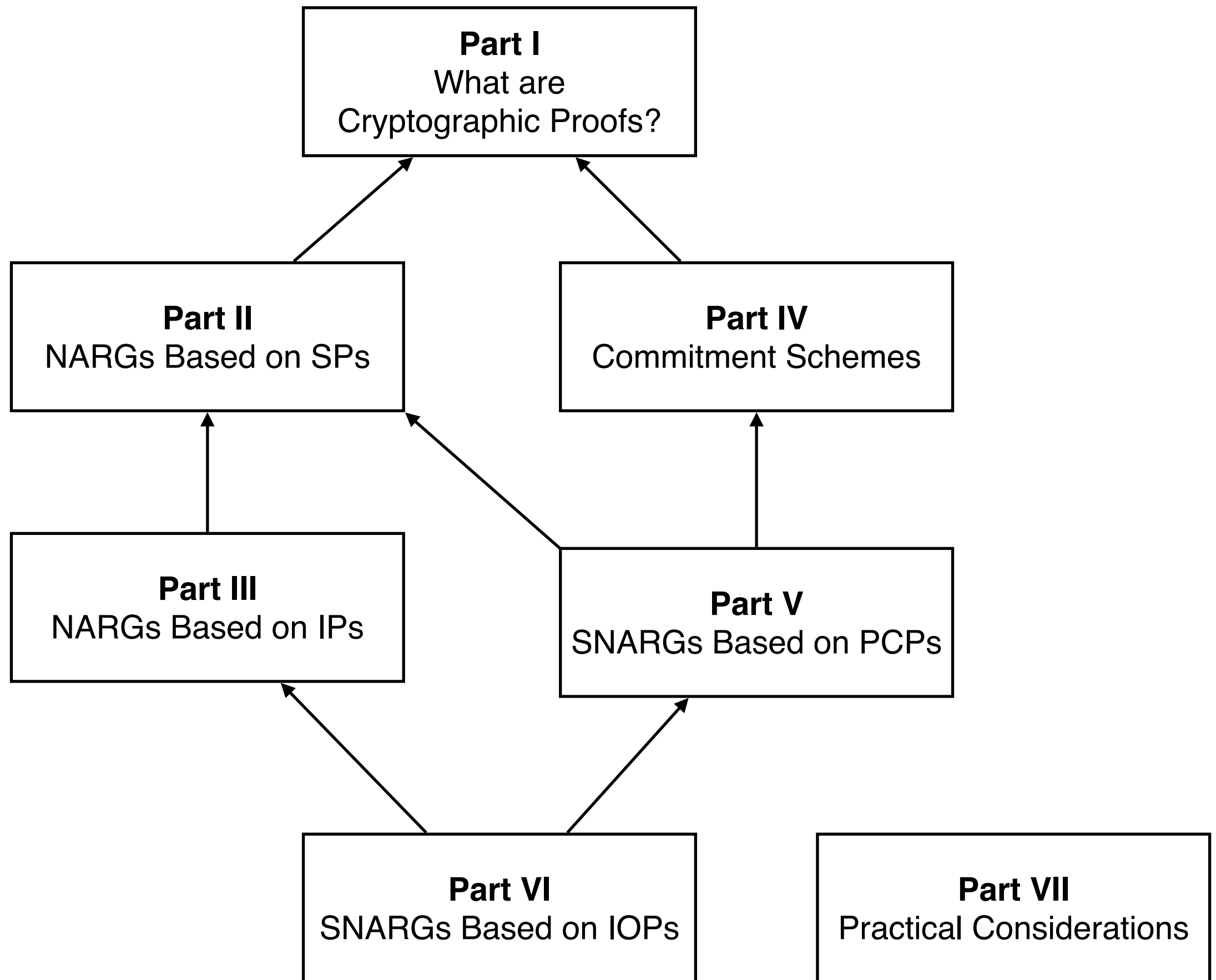
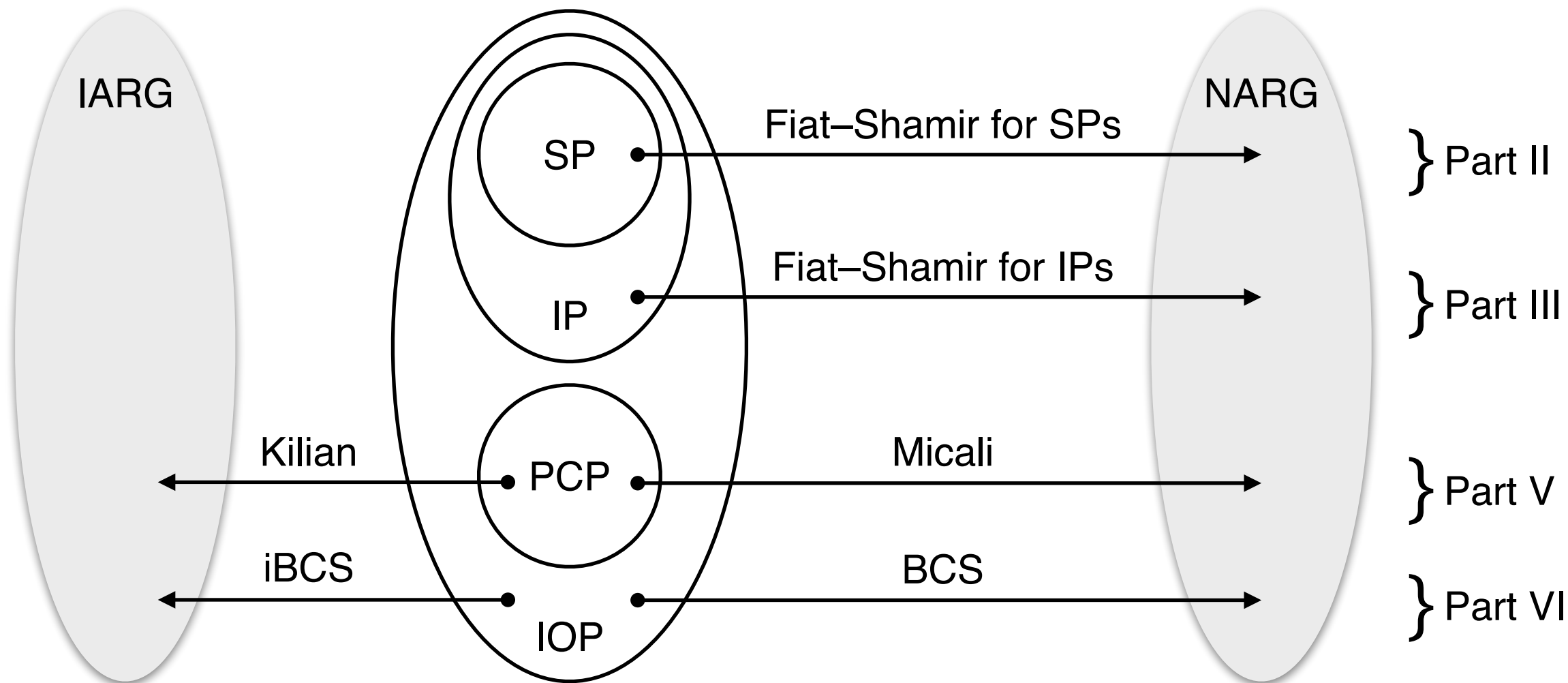
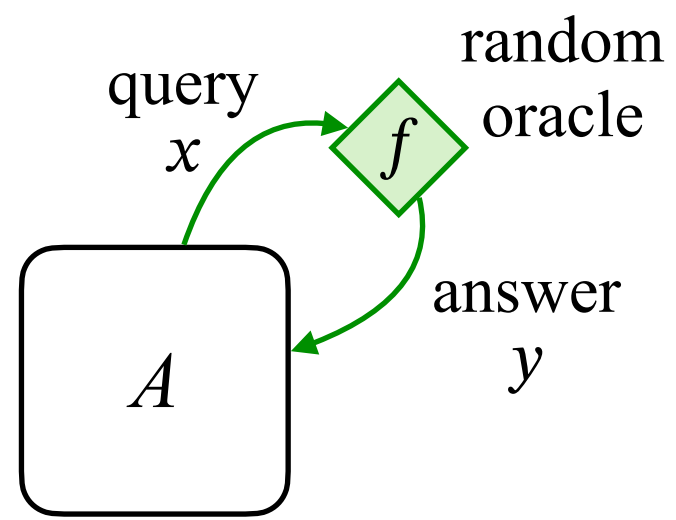
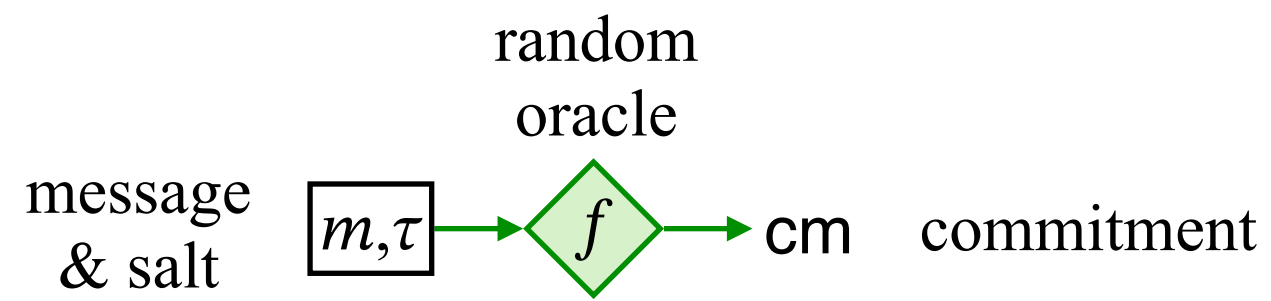
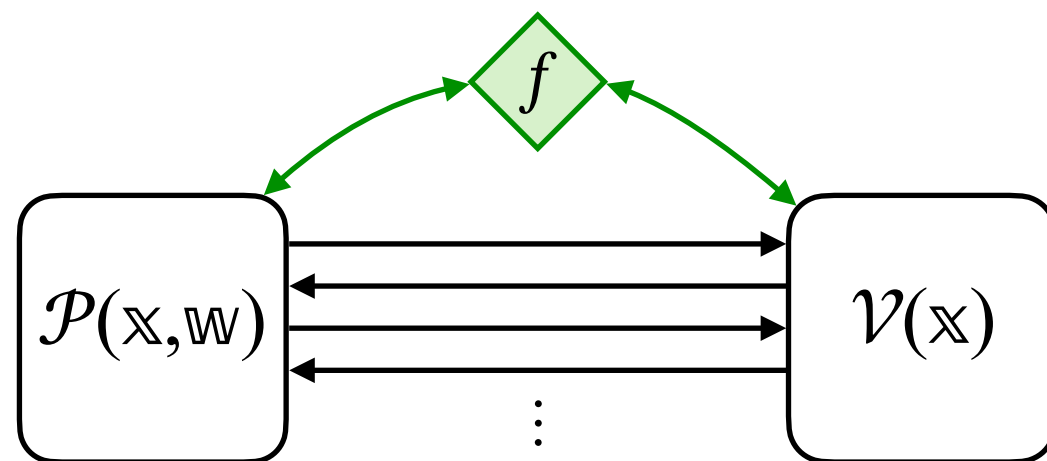


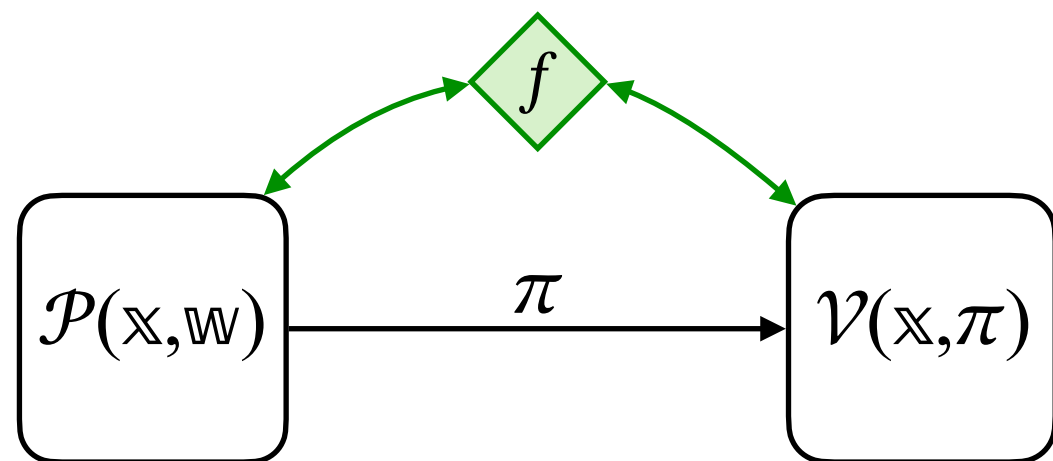


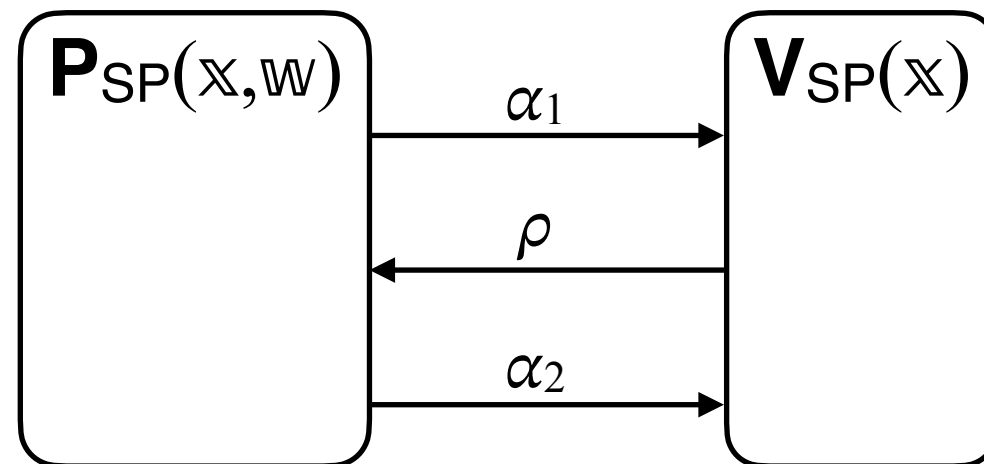


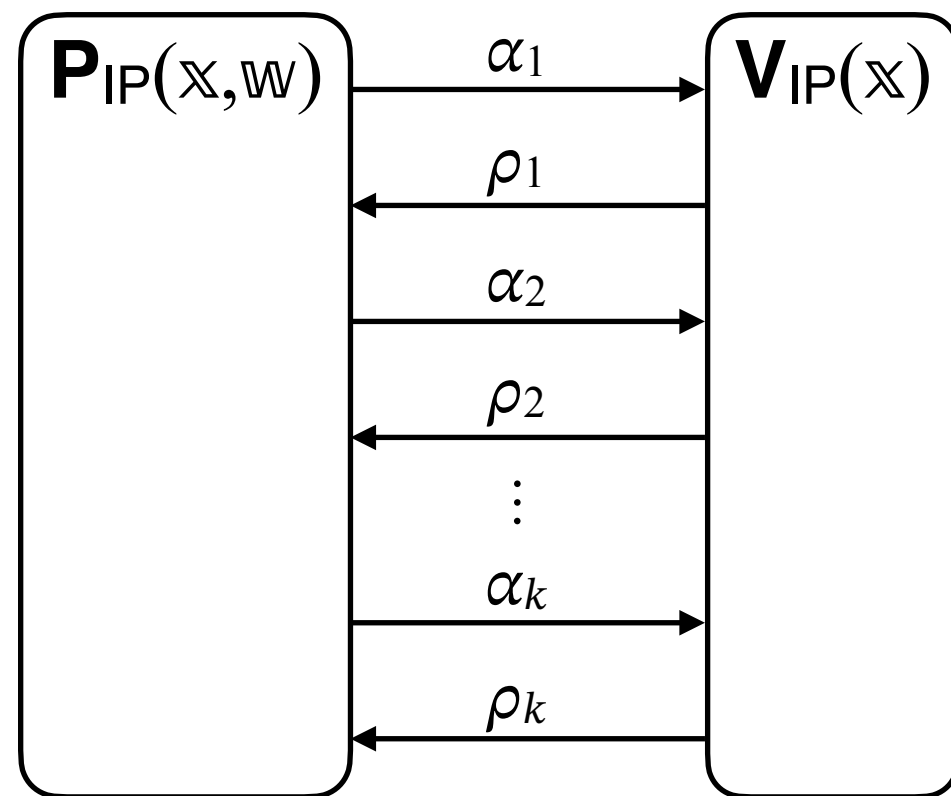


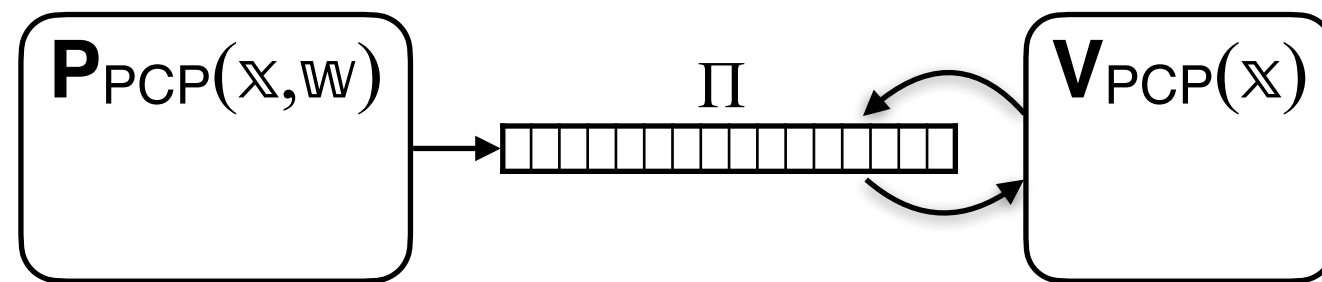


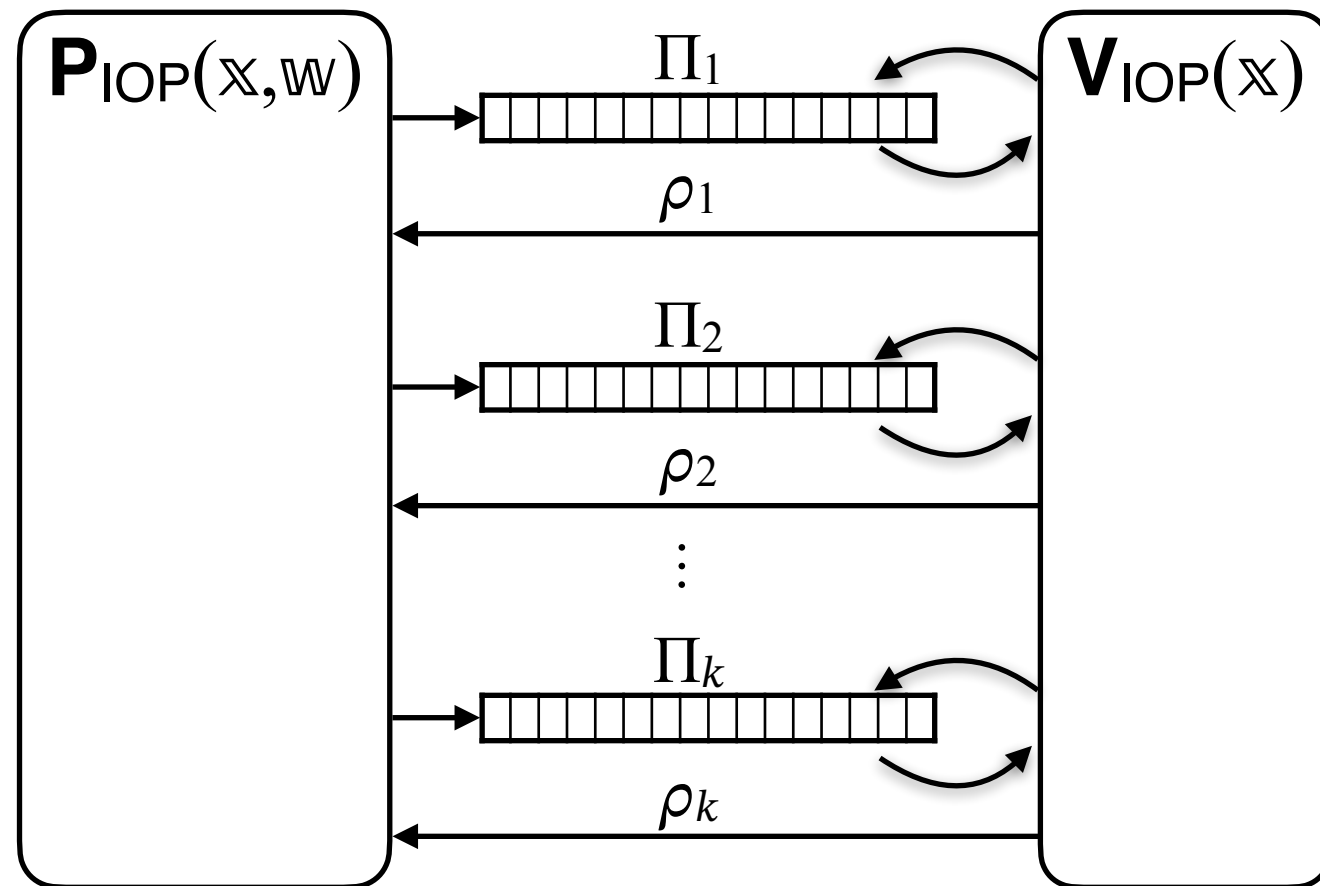


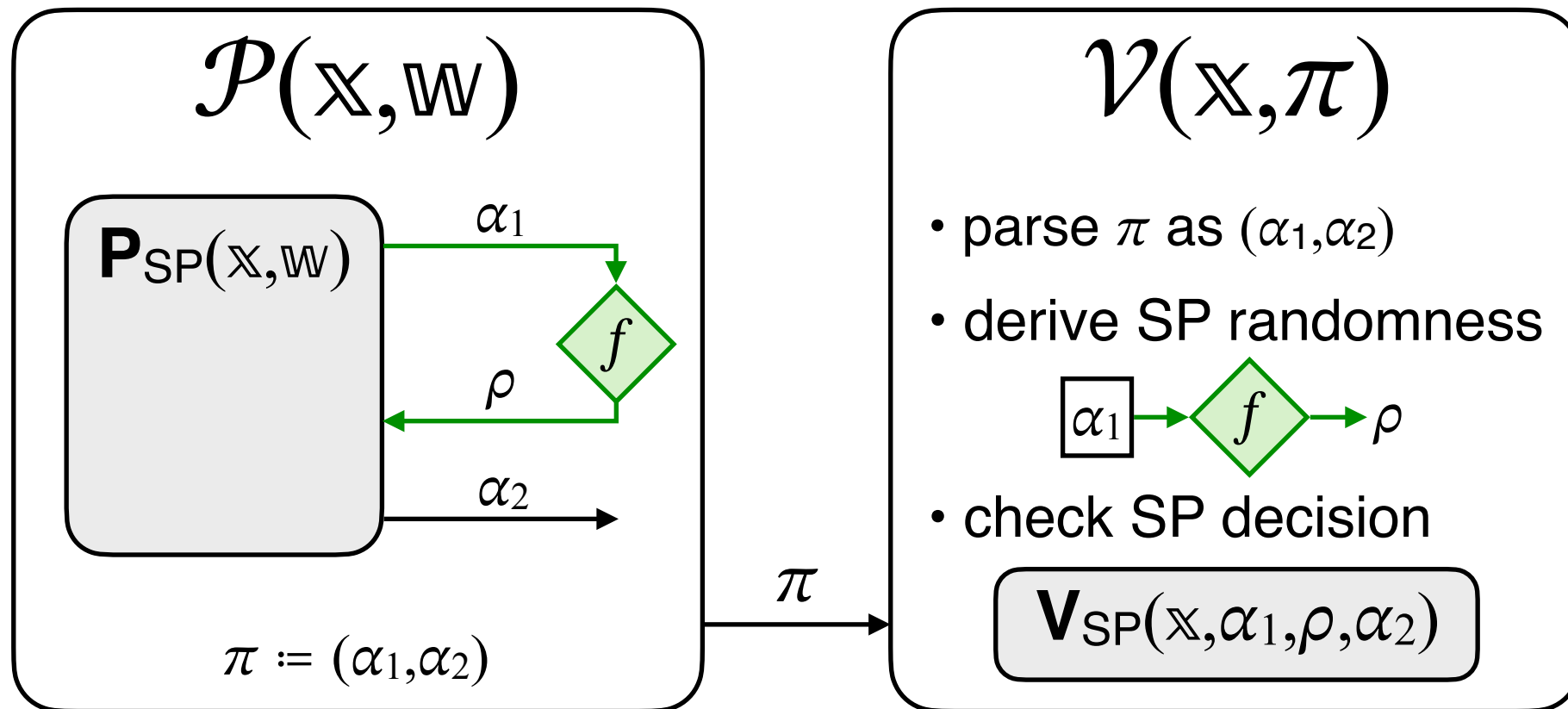


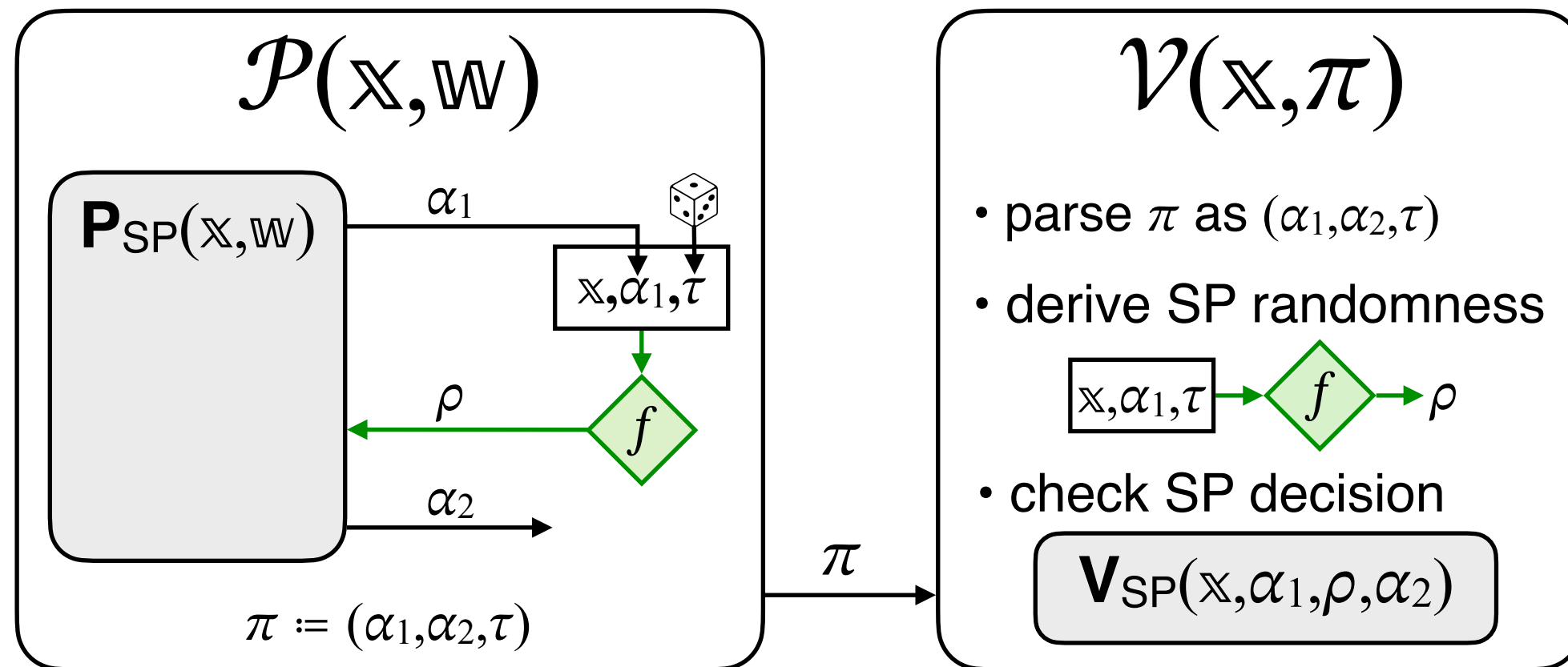




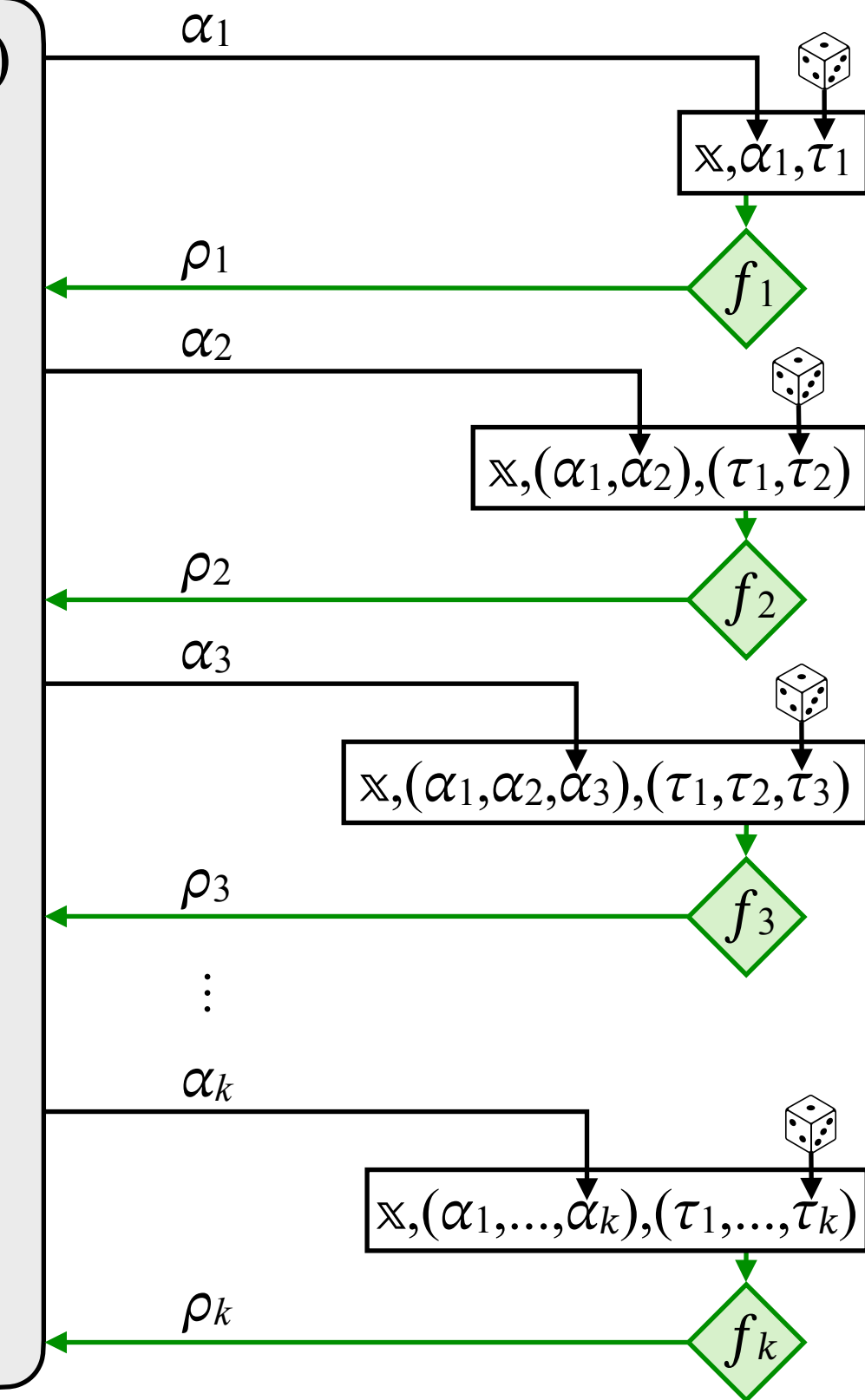








$$\mathcal{P}(\mathbb{X}, \mathbb{W})$$

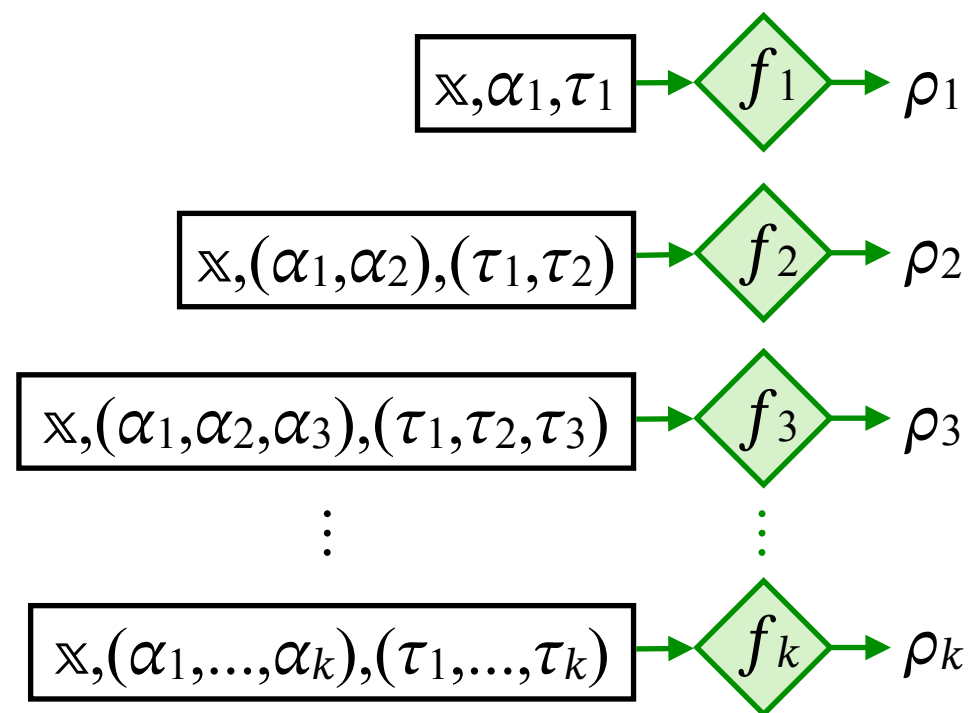
$$\mathbf{P}_{\text{IP}}(\mathbb{X}, \mathbb{W})$$


$$\pi := ((\alpha_1, \dots, \alpha_k), (\tau_1, \dots, \tau_k))$$

$$\mathcal{V}(\mathbb{X}, \pi)$$

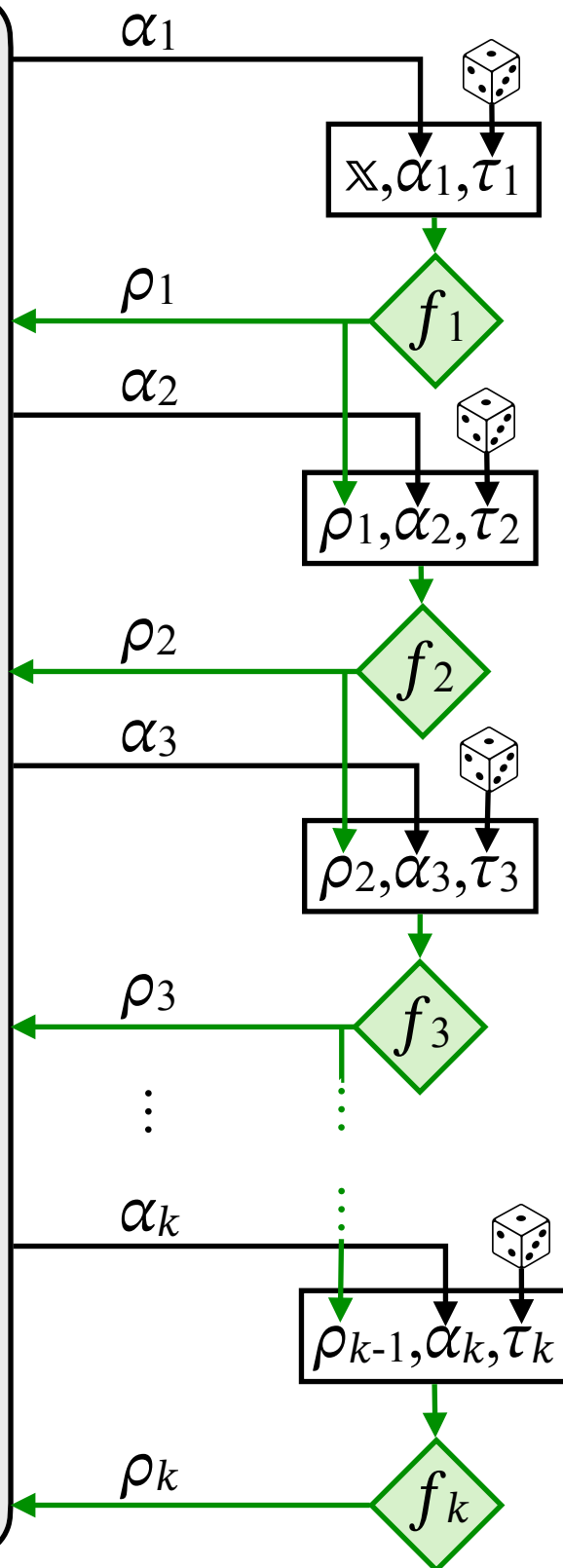
- parse π as $((\alpha_1, \dots, \alpha_k), (\tau_1, \dots, \tau_k))$

- derive IP randomness



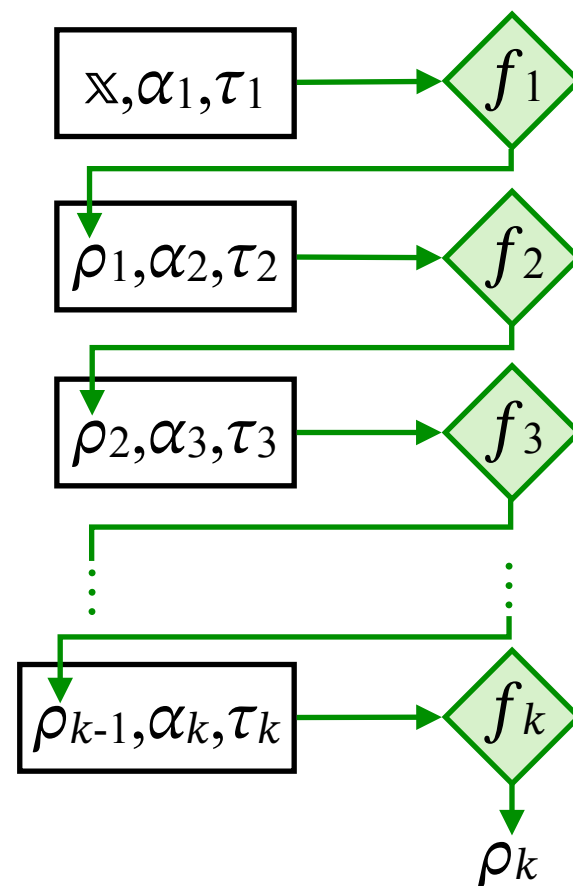
- check IP decision

$$\mathbf{V}_{\text{IP}}(\mathbb{X}, (\alpha_1, \dots, \alpha_k), (\rho_1, \dots, \rho_k))$$

$\mathcal{P}(\mathbb{X}, \mathbb{W})$
 $\mathbf{P}_{\text{IP}}(\mathbb{X}, \mathbb{W})$

 $\pi := ((\alpha_1, \dots, \alpha_k), (\tau_1, \dots, \tau_k))$
 π
 $\mathcal{V}(\mathbb{X}, \pi)$

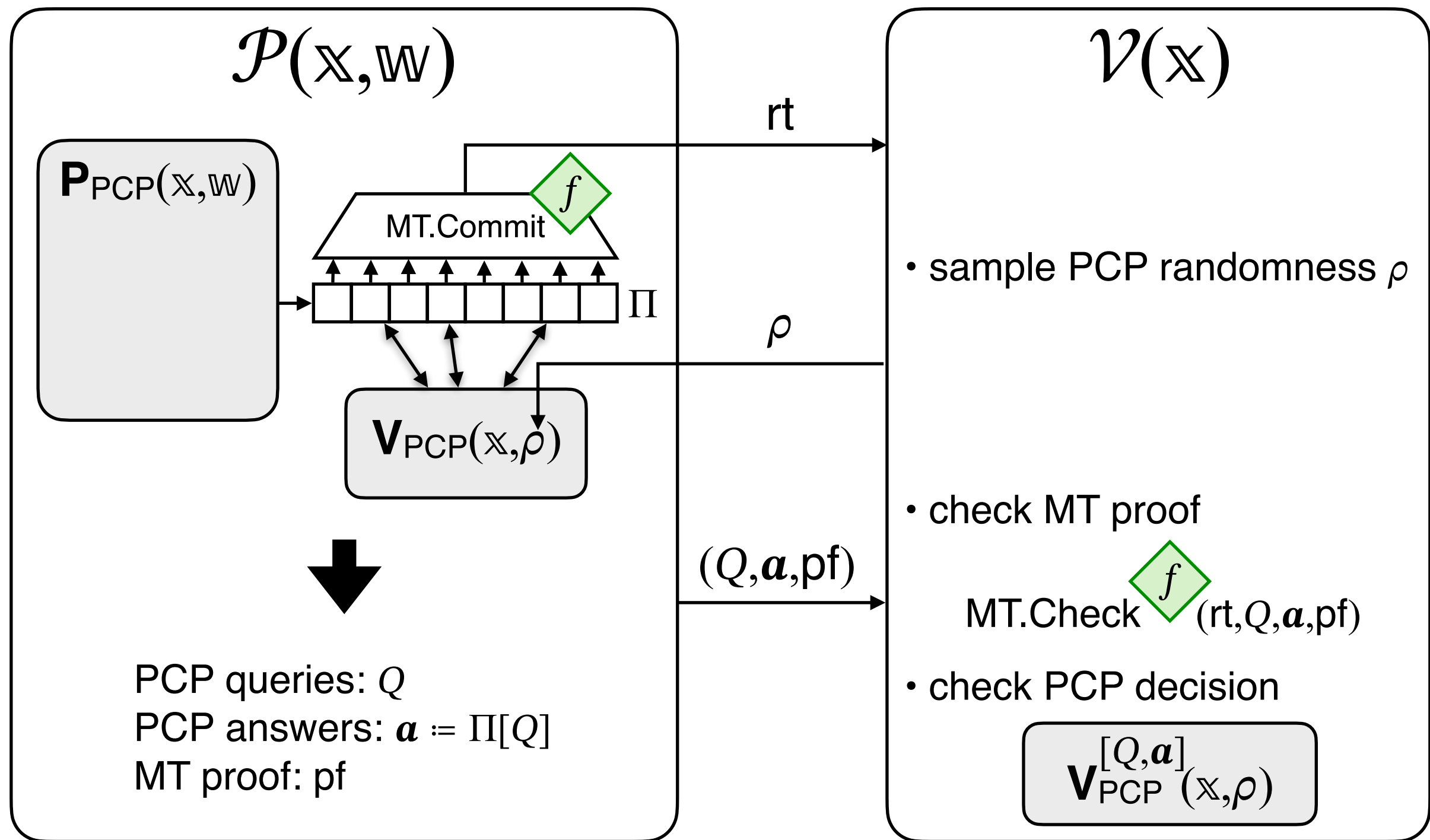
- parse π as $((\alpha_1, \dots, \alpha_k), (\tau_1, \dots, \tau_k))$

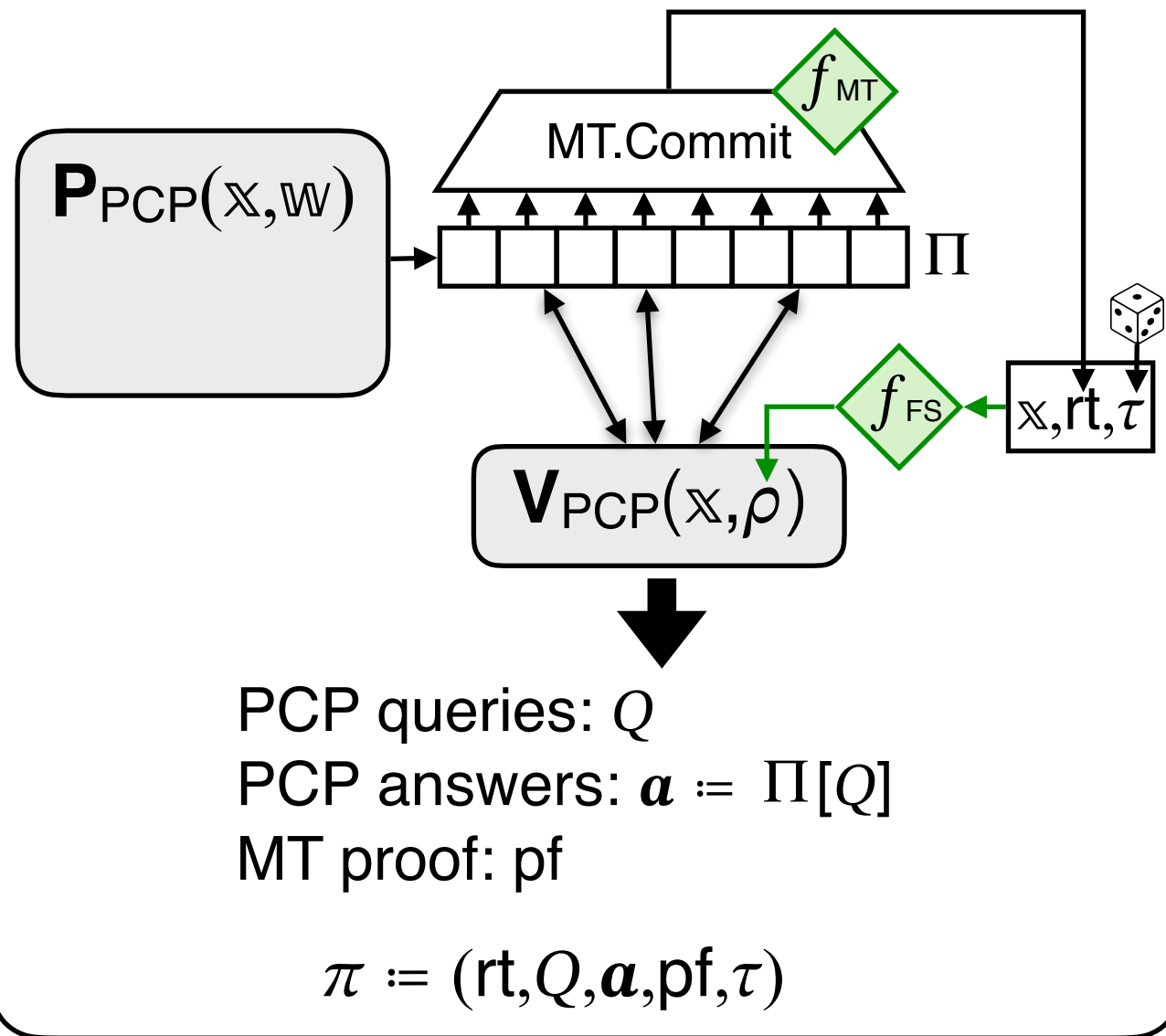
- derive IP randomness

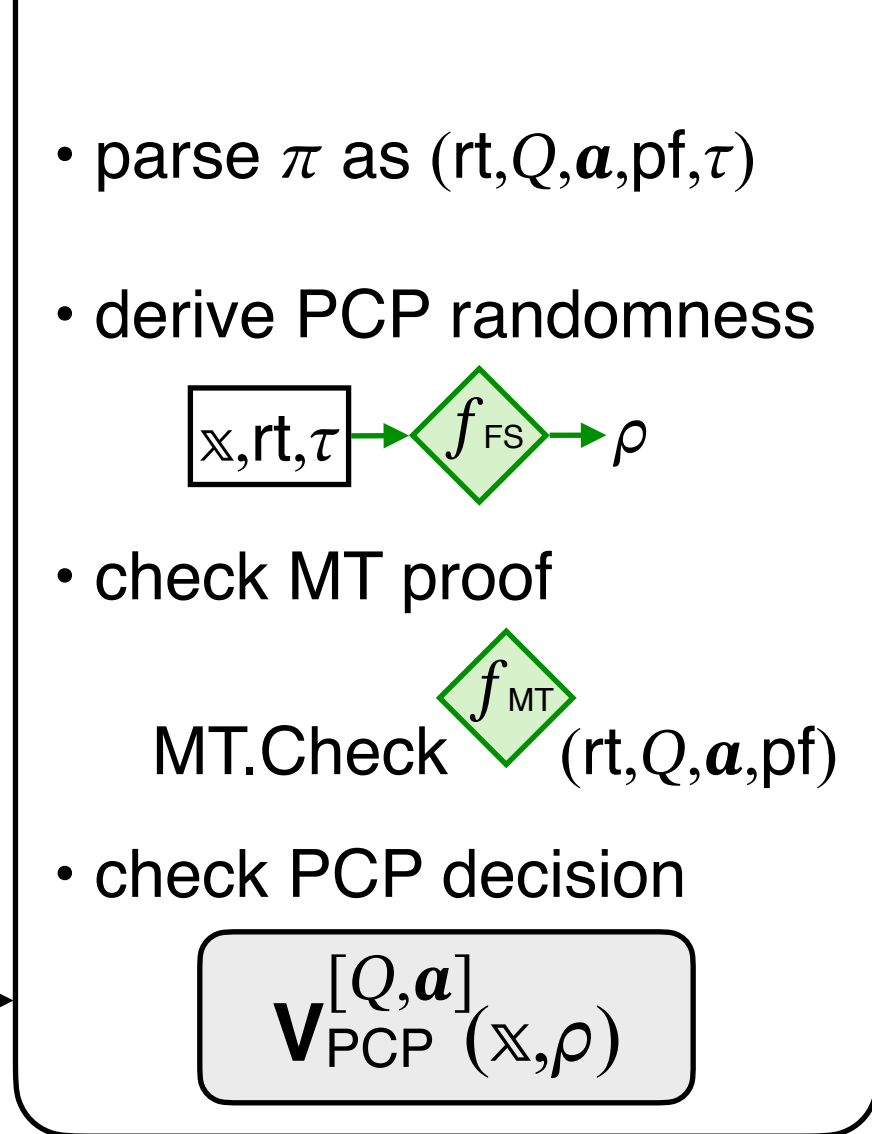


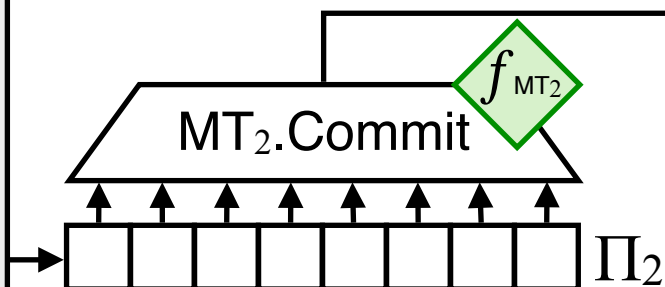
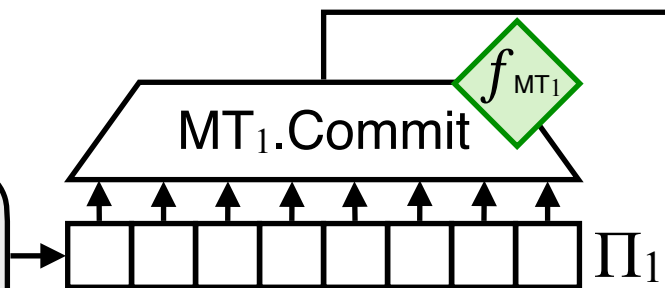
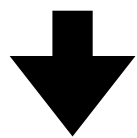
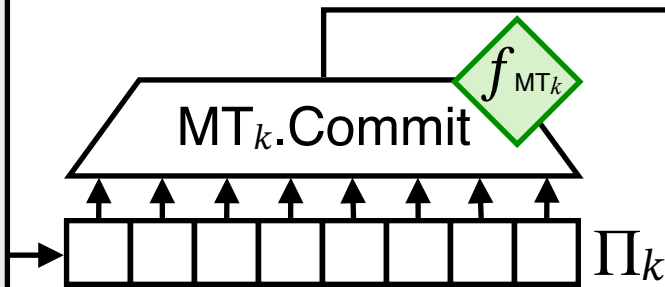
- check IP decision

 $\mathbf{V}_{\text{IP}}(\mathbb{X}, (\alpha_1, \dots, \alpha_k), (\rho_1, \dots, \rho_k))$



$$\mathcal{P}(\mathbb{X}, \mathbb{W})$$


$$\mathcal{V}(\mathbb{X}, \pi)$$


$\mathcal{P}(\mathbb{X}, \mathbb{W})$
 $\mathcal{V}(\mathbb{X})$
 $\mathbf{P}_{\text{IOP}}(\mathbb{X}, \mathbb{W})$

 $\vdots$


IOP verifier queries: $(Q_1, \dots, Q_k)$
 IOP oracle answers: $(\mathbf{a}_1, \dots, \mathbf{a}_k)$
 MT proofs: $(\text{pf}_1, \dots, \text{pf}_k)$

 rt_1
 ρ_1
 rt_2
 ρ_2
 $\vdots$
 rt_k
 ρ_k

- sample IOP randomness ρ_1

- sample IOP randomness ρ_2

 $\vdots$

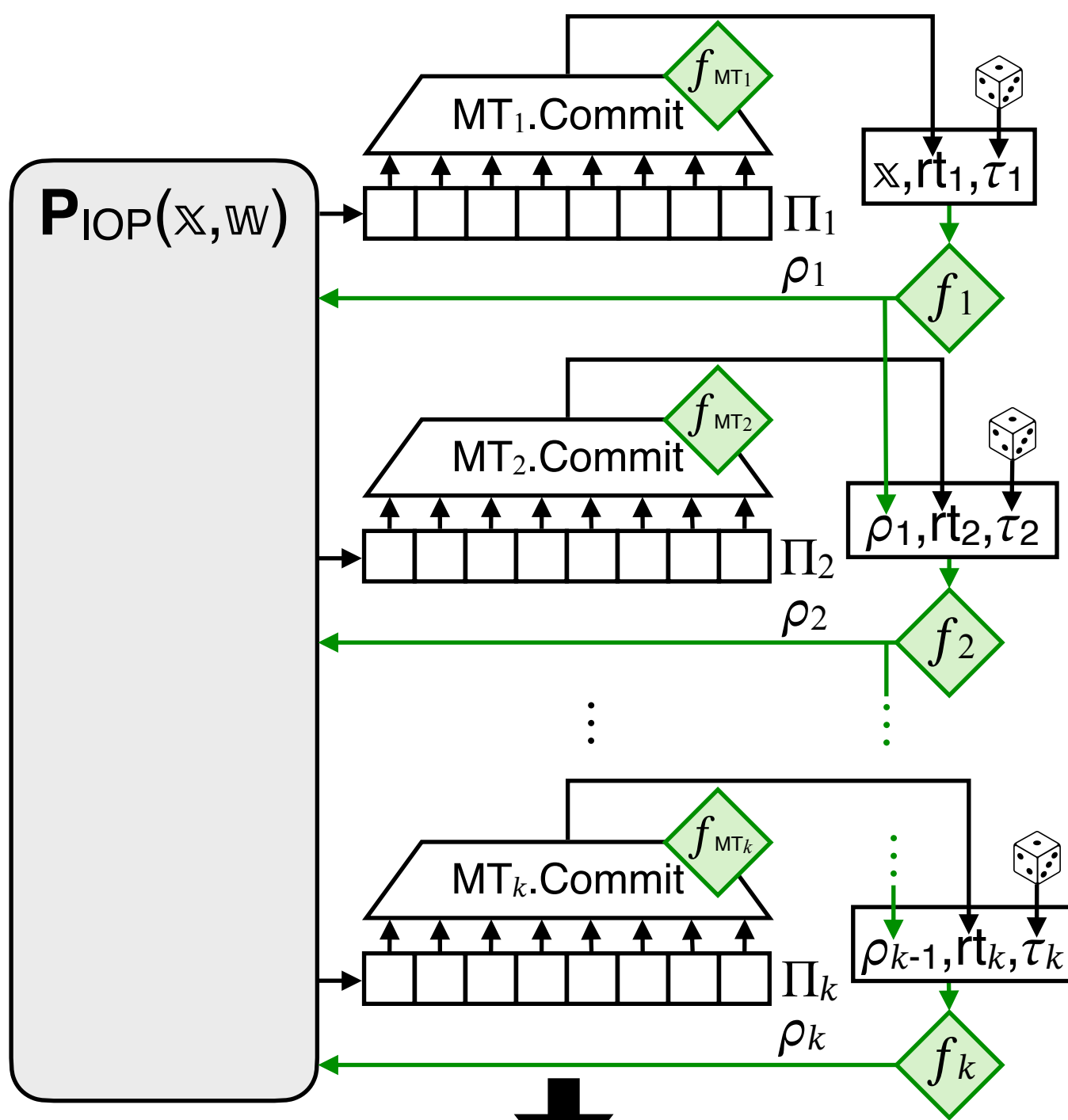
- sample IOP randomness ρ_k

- check MT proofs

 $\bigwedge_{i \in [k]} \text{MT}_i.\text{Check}^{f_{\text{MT}_i}}(\text{rt}_i, Q_i, \mathbf{a}_i, \text{pf}_i)$

- check IOP decision

$$\mathbf{V}_{\text{IOP}}^{[Q_i, \mathbf{a}_i]_{i \in [k]}}(\mathbb{X}, (\rho_1, \dots, \rho_k))$$
 $((Q_i, \mathbf{a}_i, \text{pf}_i))_{i \in [k]}$

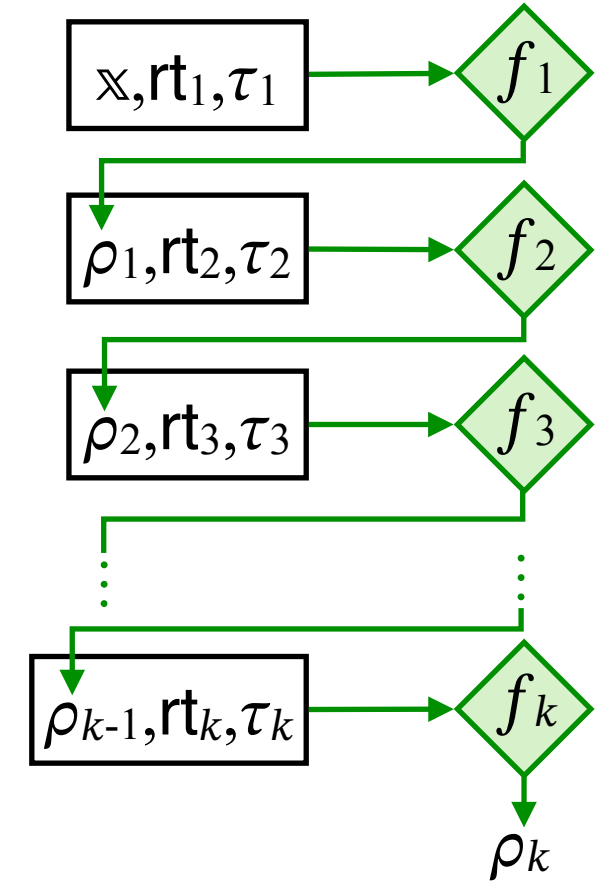
$\mathcal{P}(\mathbb{X}, \mathbb{W})$


IOP verifier queries: $(Q_1, \dots, Q_k)$
 IOP oracle answers: $(\mathbf{a}_1, \dots, \mathbf{a}_k)$
 MT proofs: $(\text{pf}_1, \dots, \text{pf}_k)$
 $\pi := ((rt_i, Q_i, \mathbf{a}_i, \text{pf}_i, \tau_i))_{i \in [k]}$

 $\mathcal{V}(\mathbb{X}, \pi)$

- parse π as $((rt_i, Q_i, \mathbf{a}_i, \text{pf}_i, \tau_i))_{i \in [k]}$

- derive IOP randomness

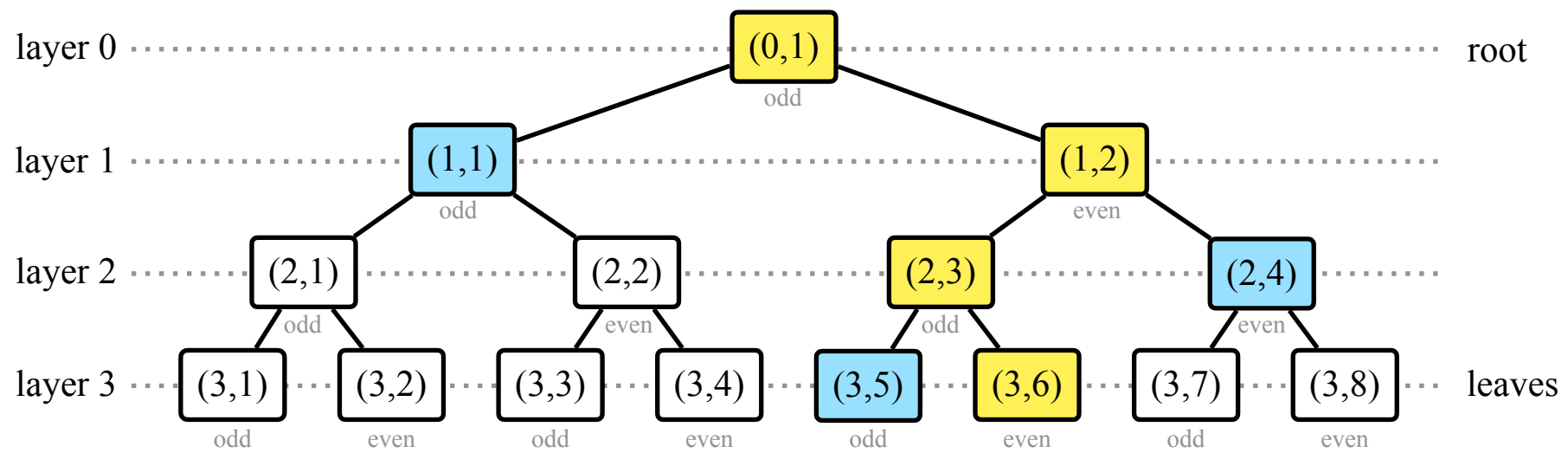


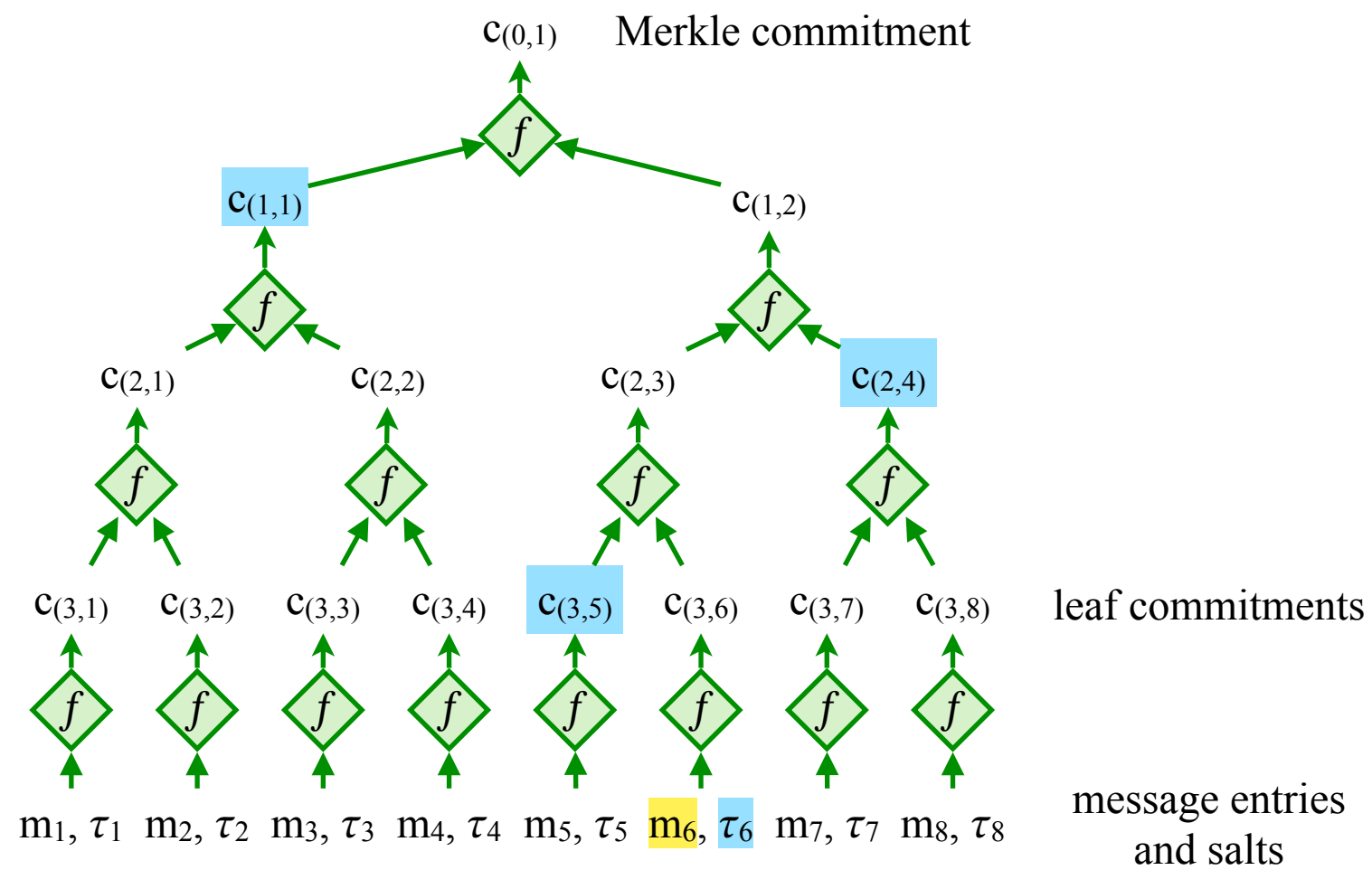
- check MT proofs

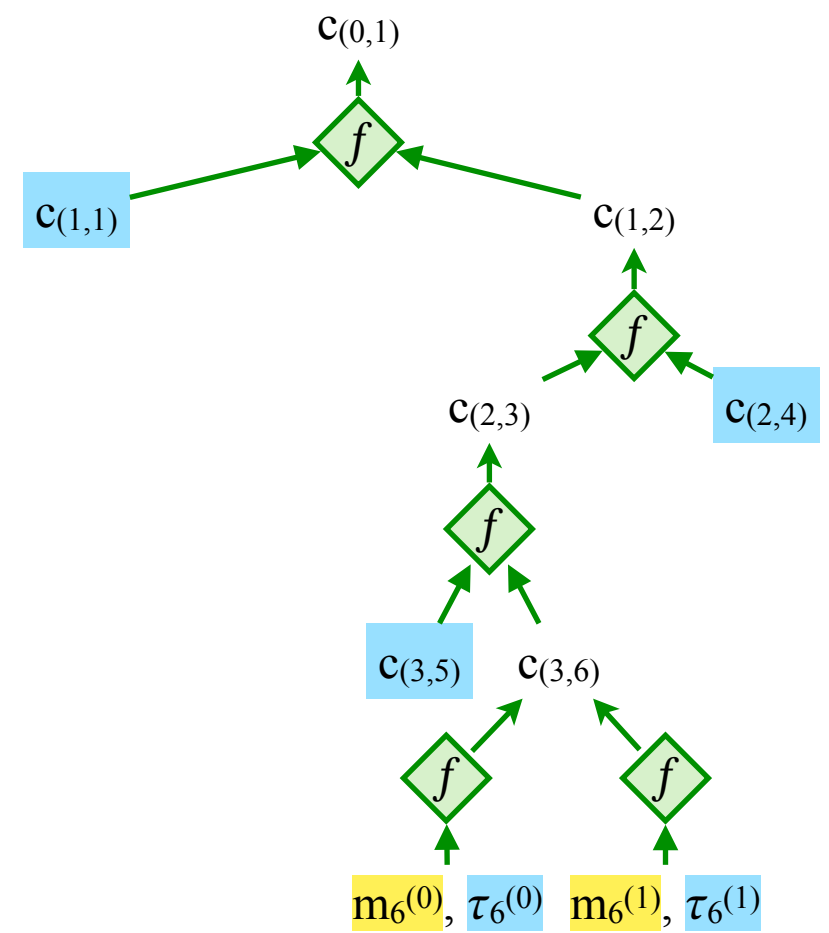
$\bigwedge_{i \in [k]} \text{MT}_i.\text{Check} \diamond f_{\text{MT}_i} (rt_i, Q_i, \mathbf{a}_i, \text{pf}_i)$

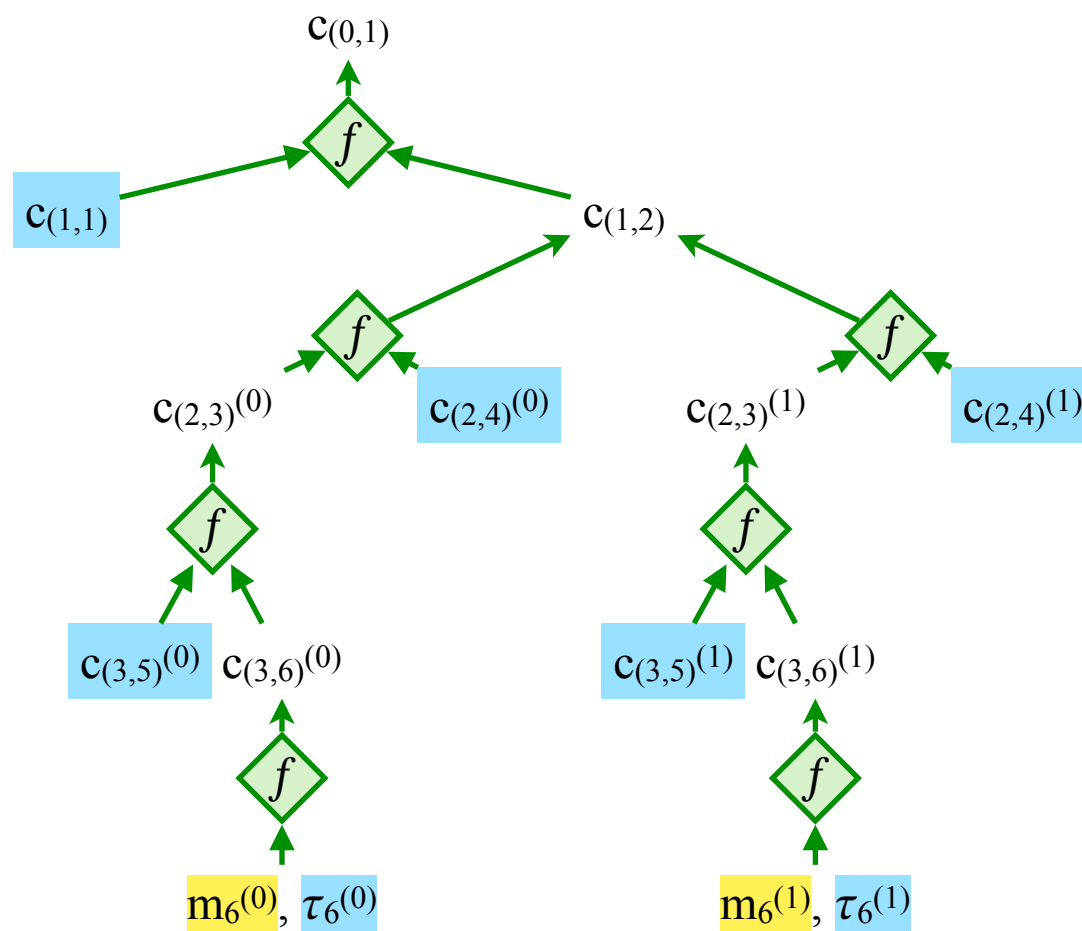
- check IOP decision

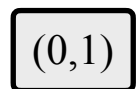
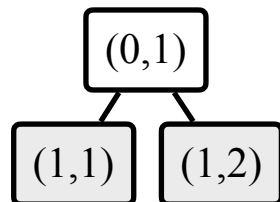
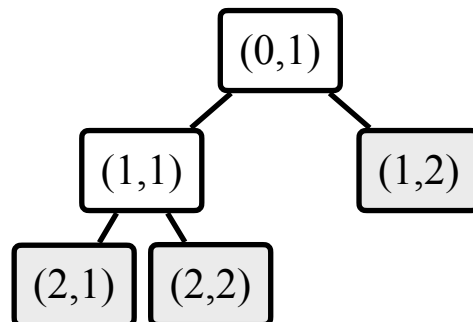
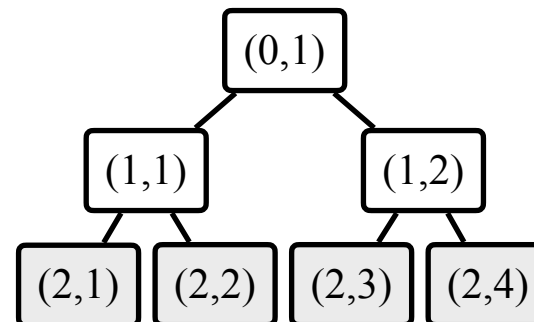
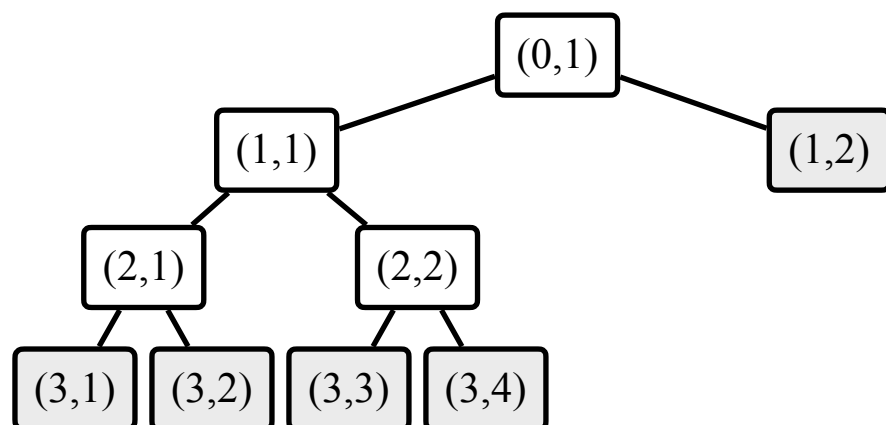
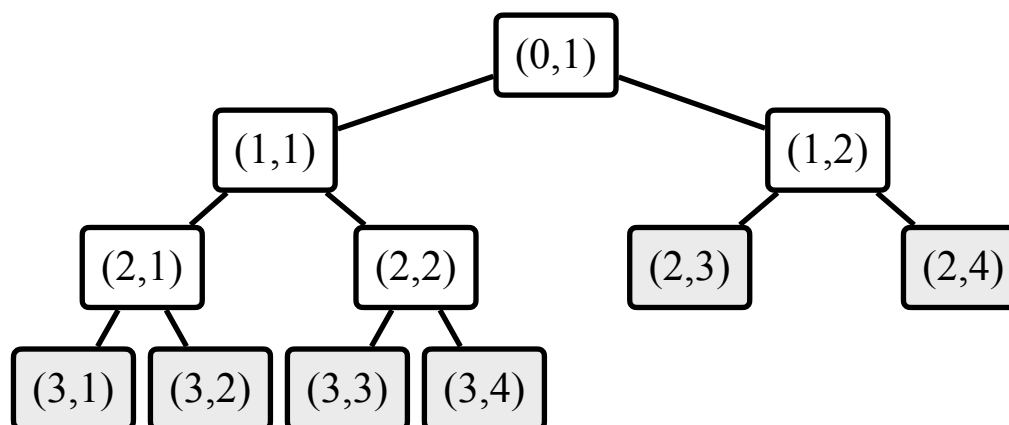
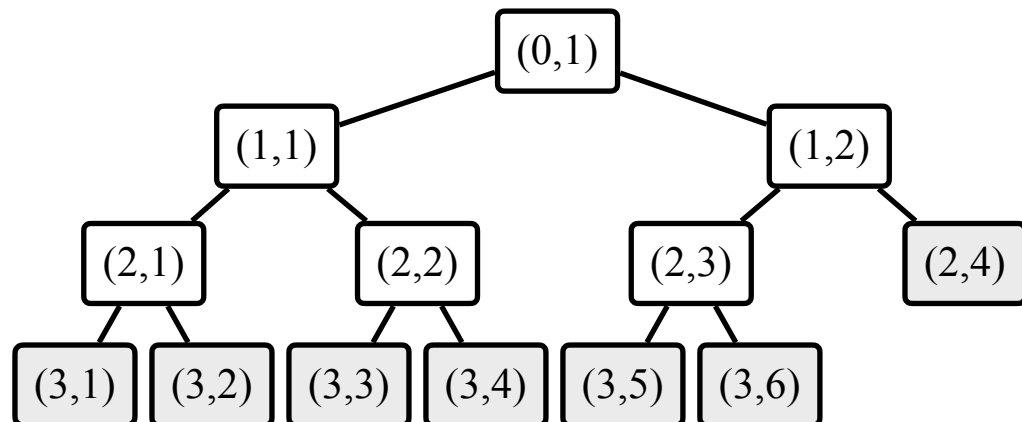
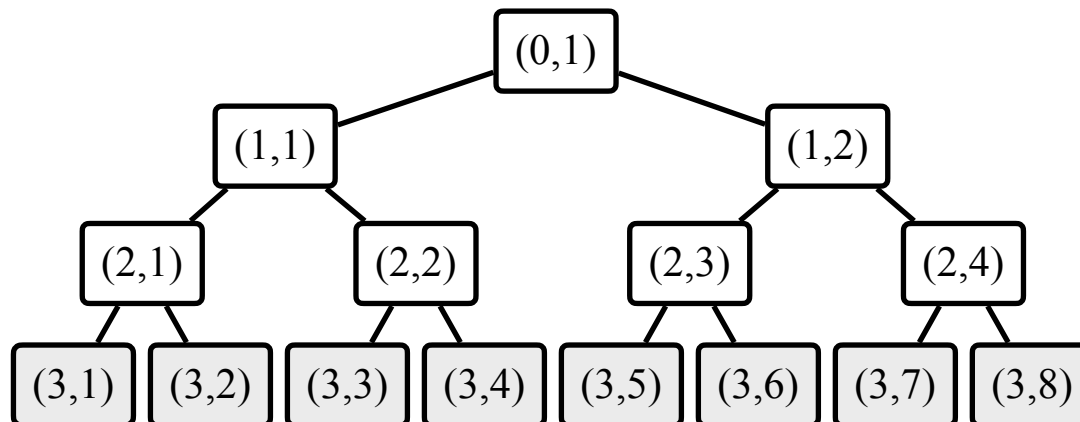
$\mathbf{V}_{\text{IOP}}^{[Q_i, \mathbf{a}_i]_{i \in [k]}}(\mathbb{X}, (\rho_1, \dots, \rho_k))$

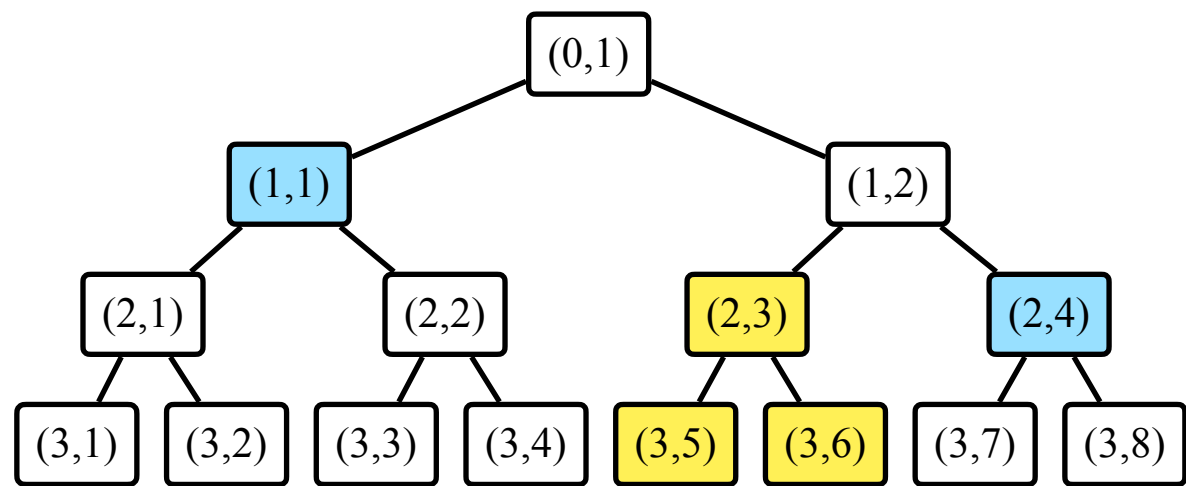


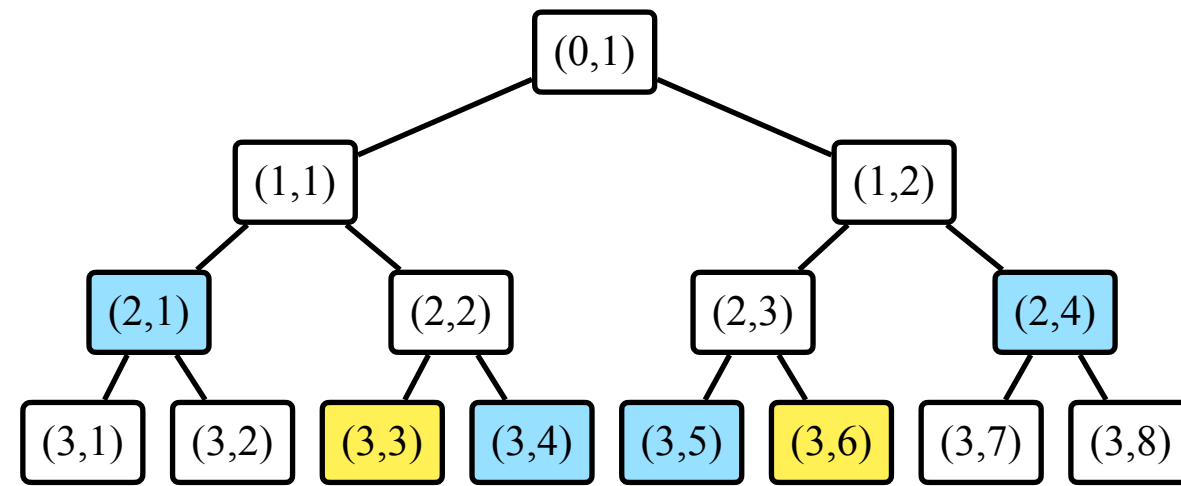


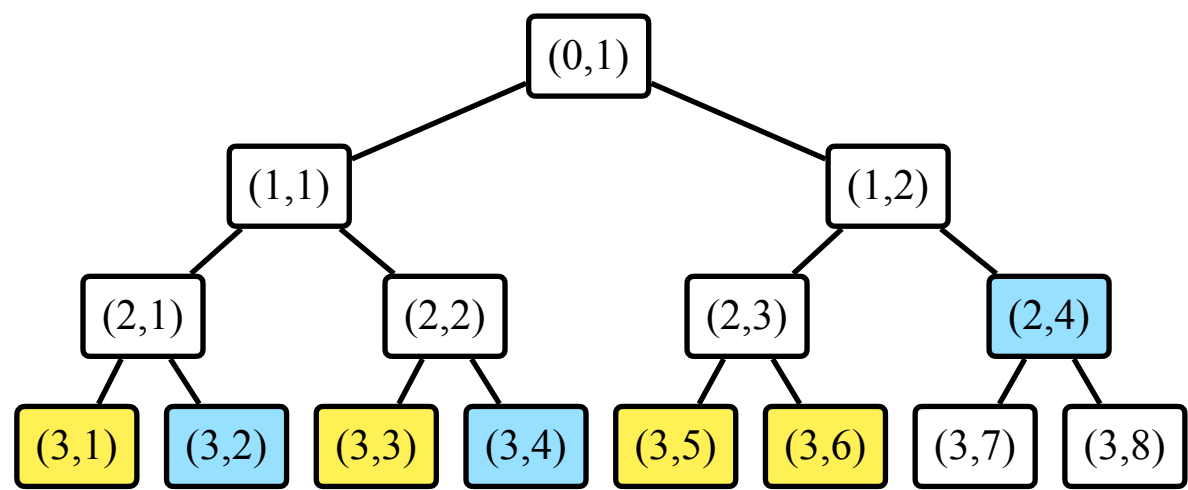


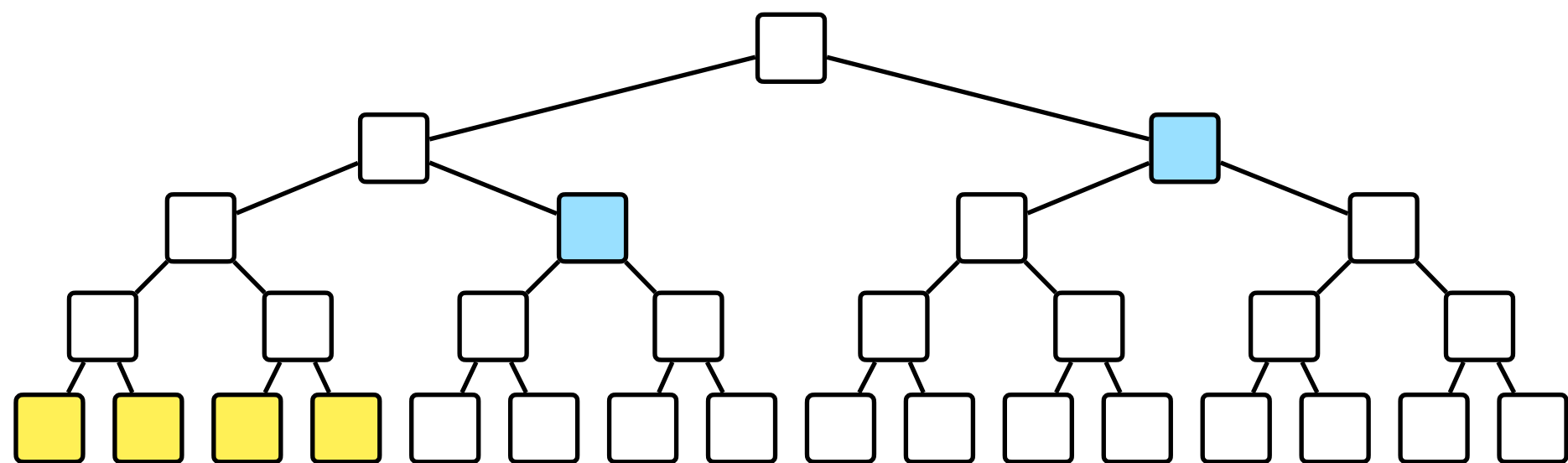


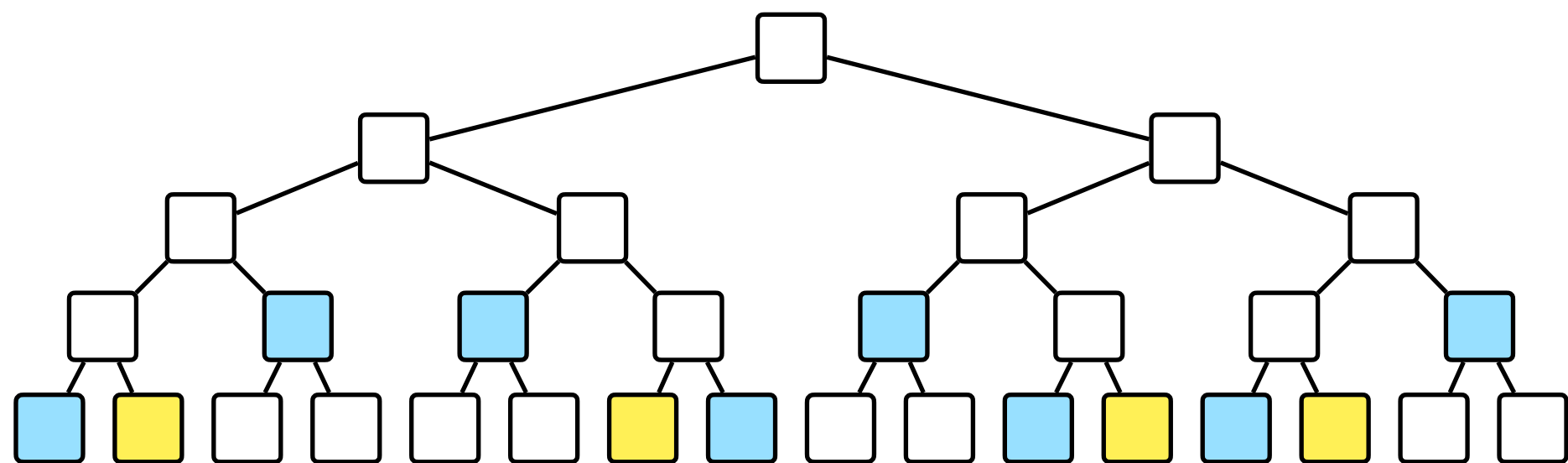
T_1  T_2  T_3  T_4  T_5  T_6  T_7  T_8 

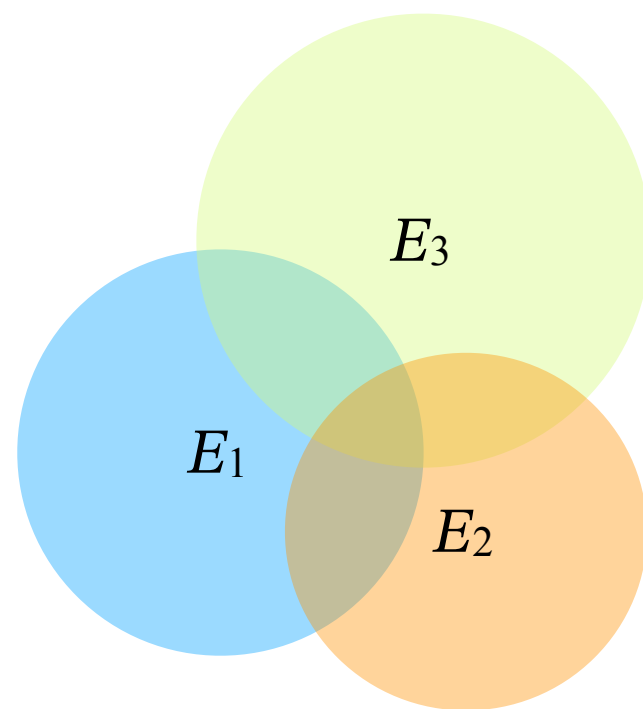


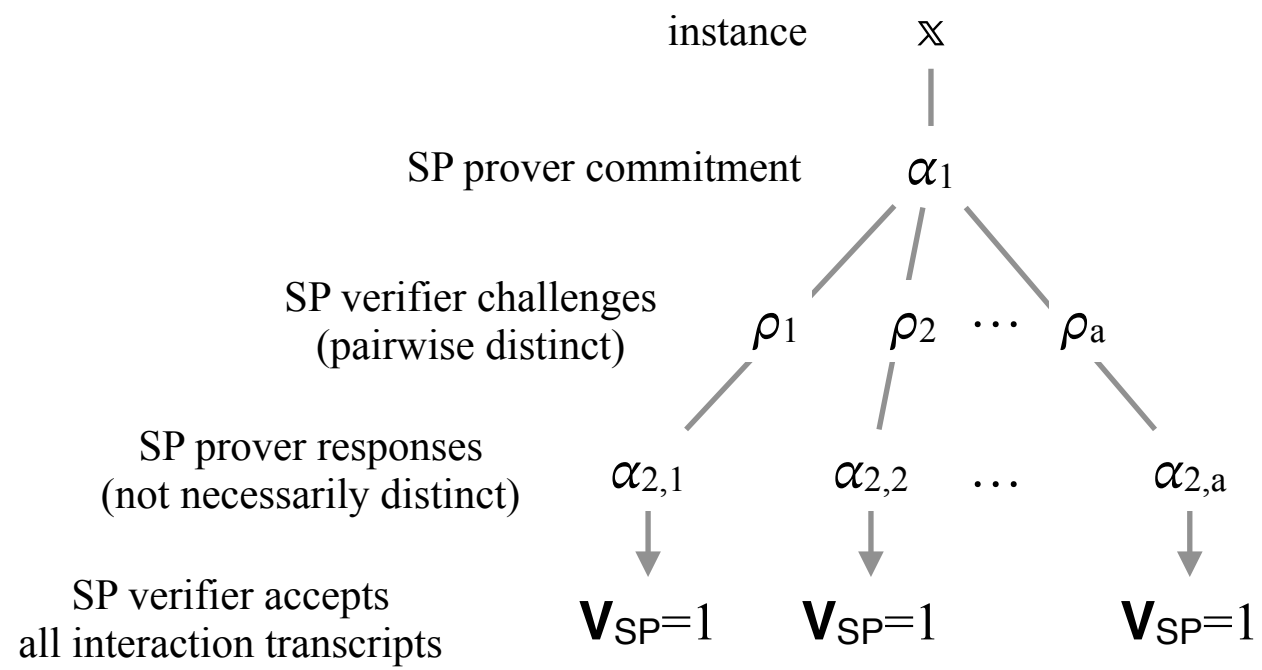


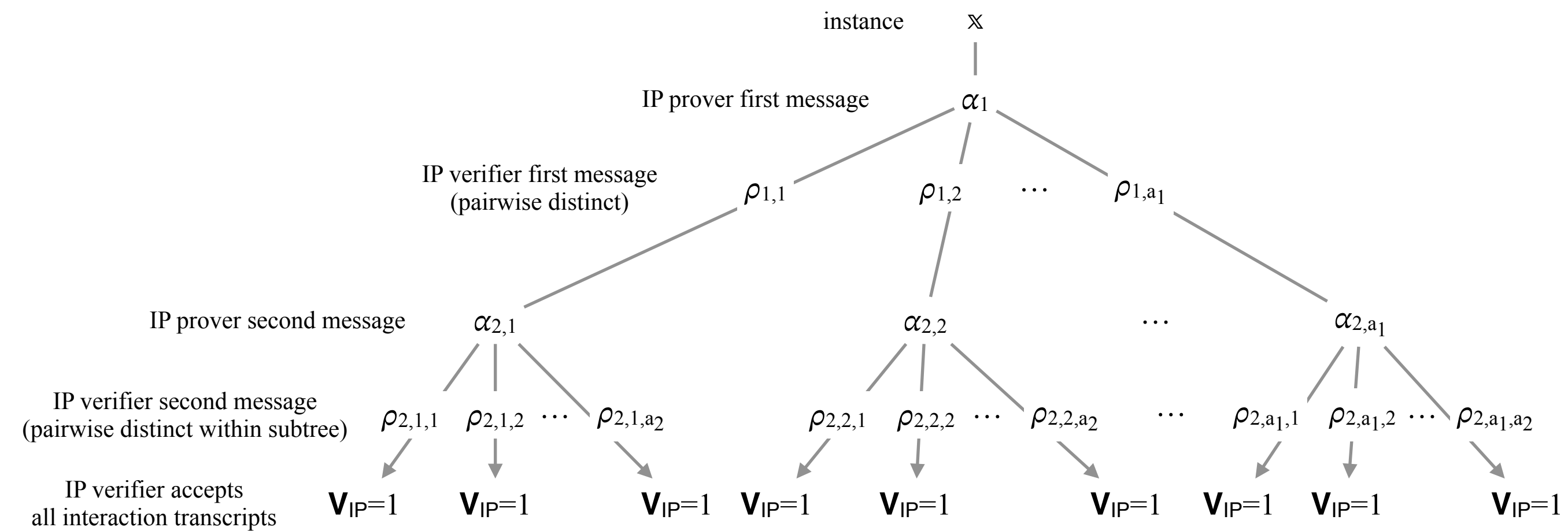


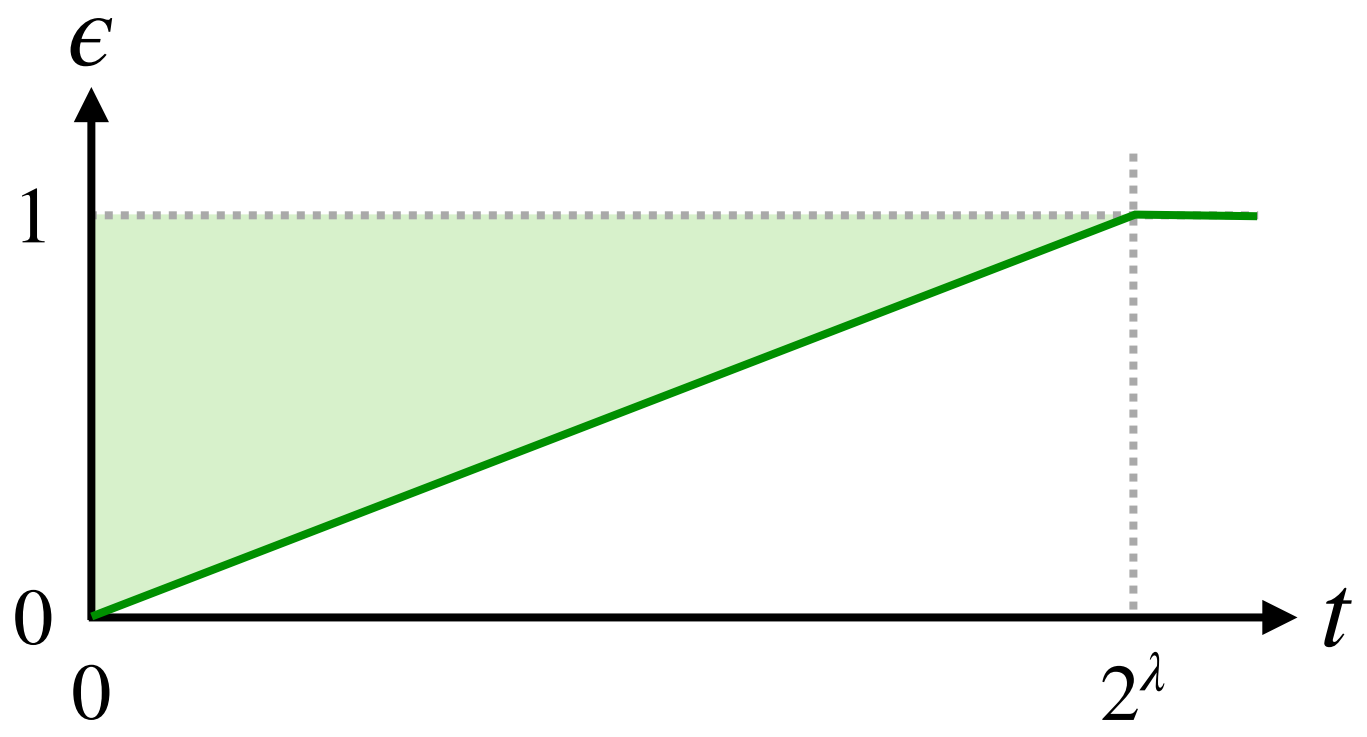


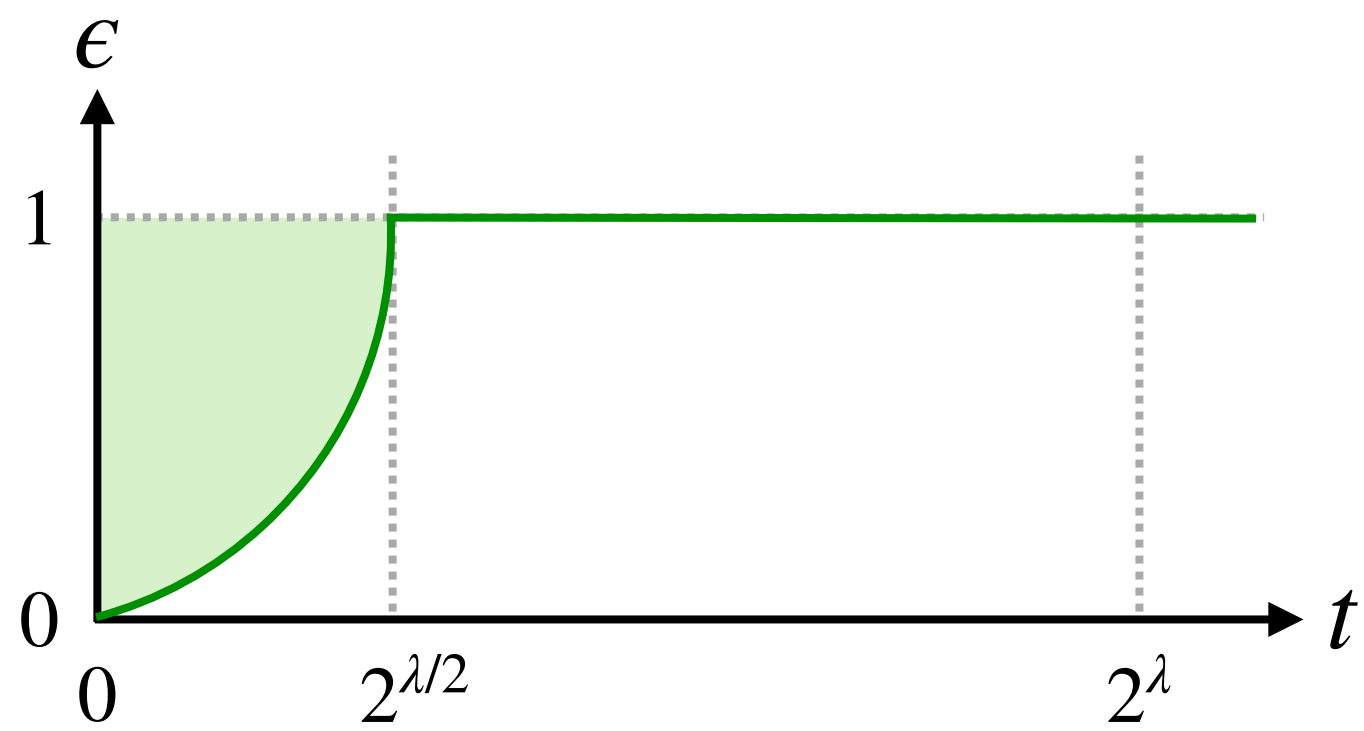


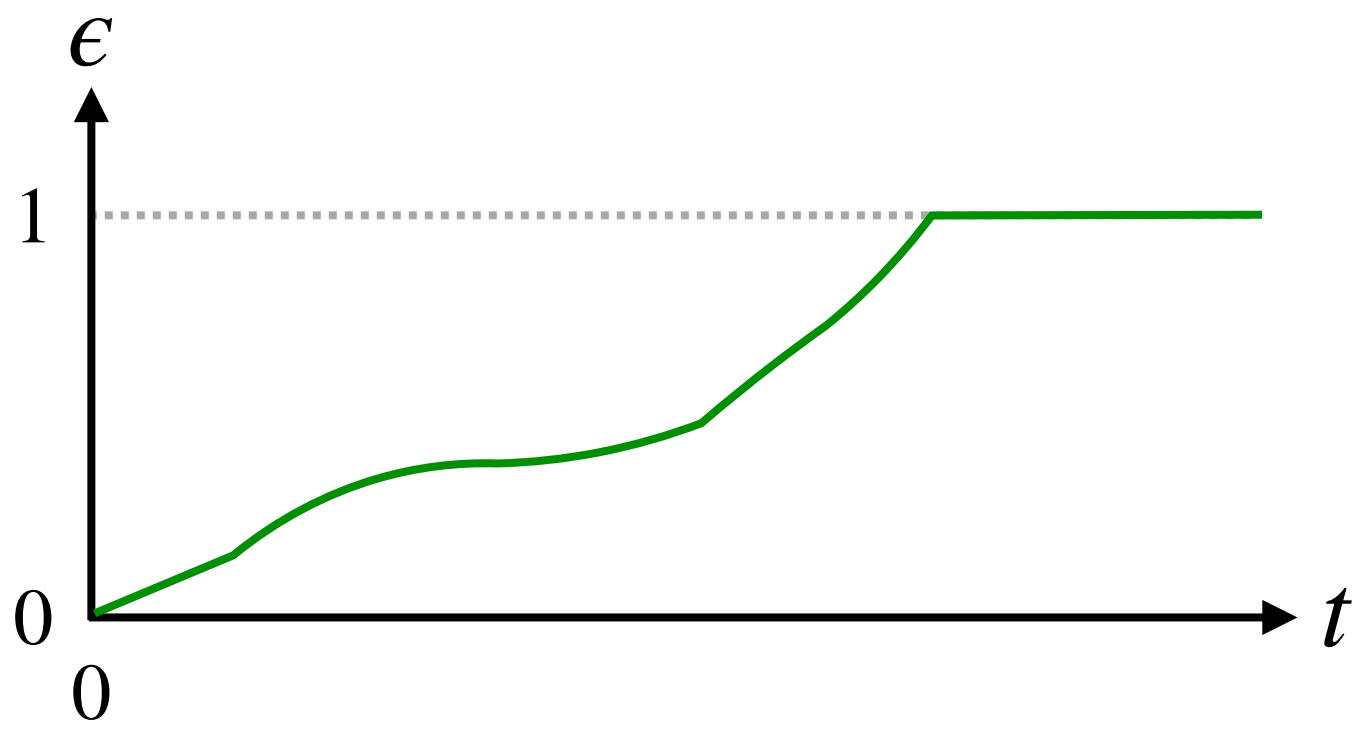


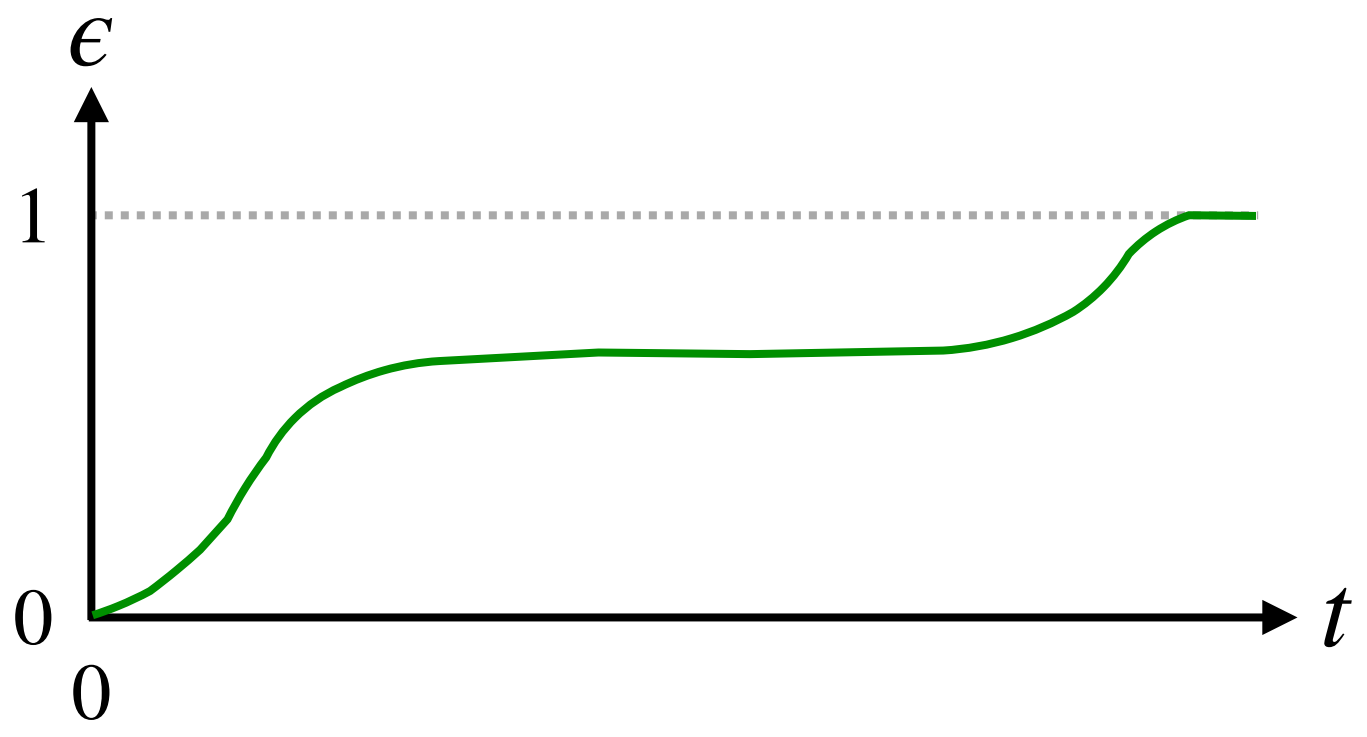


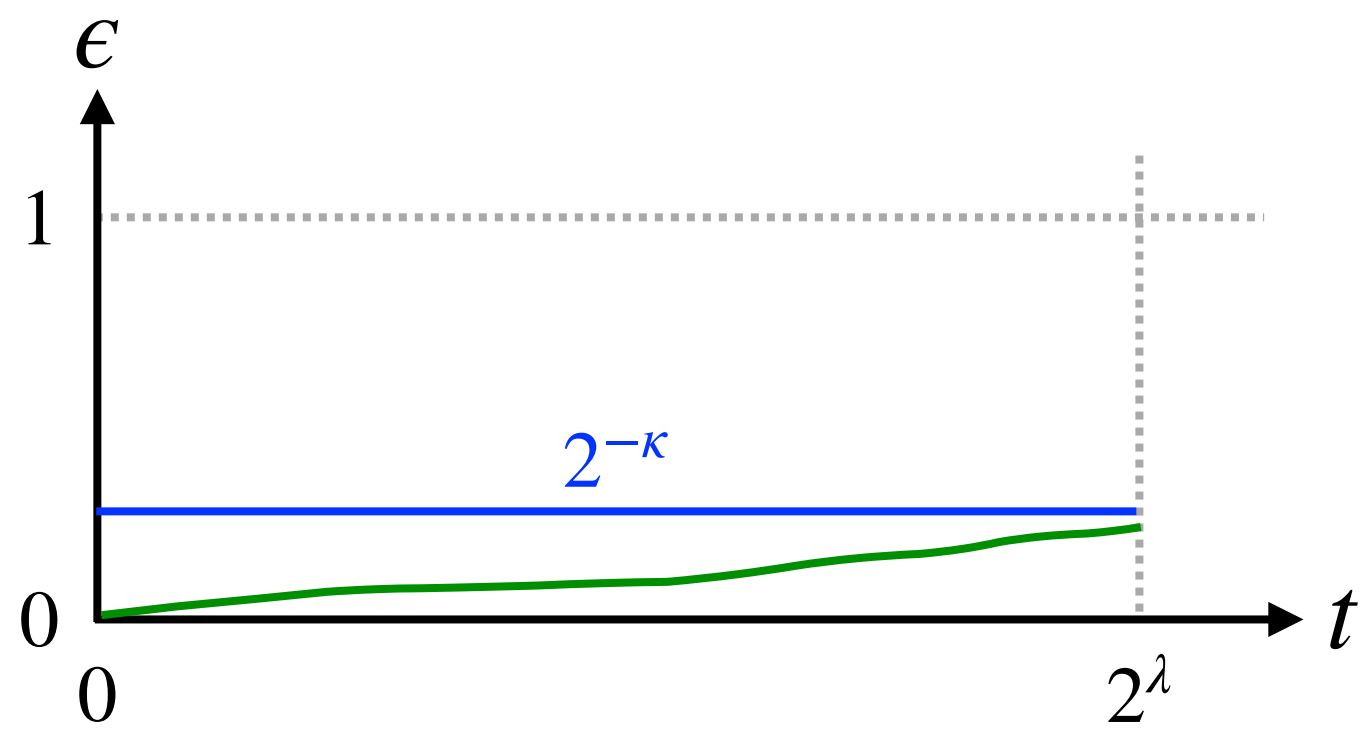


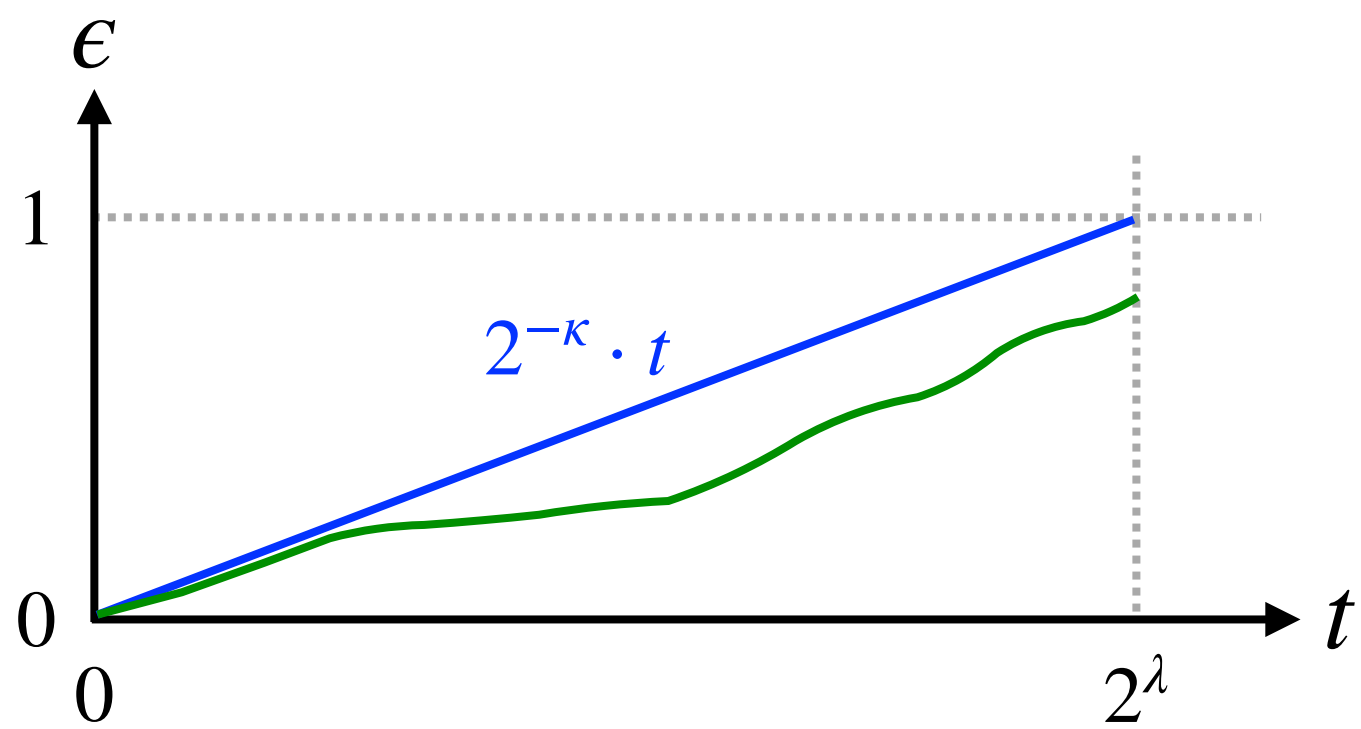


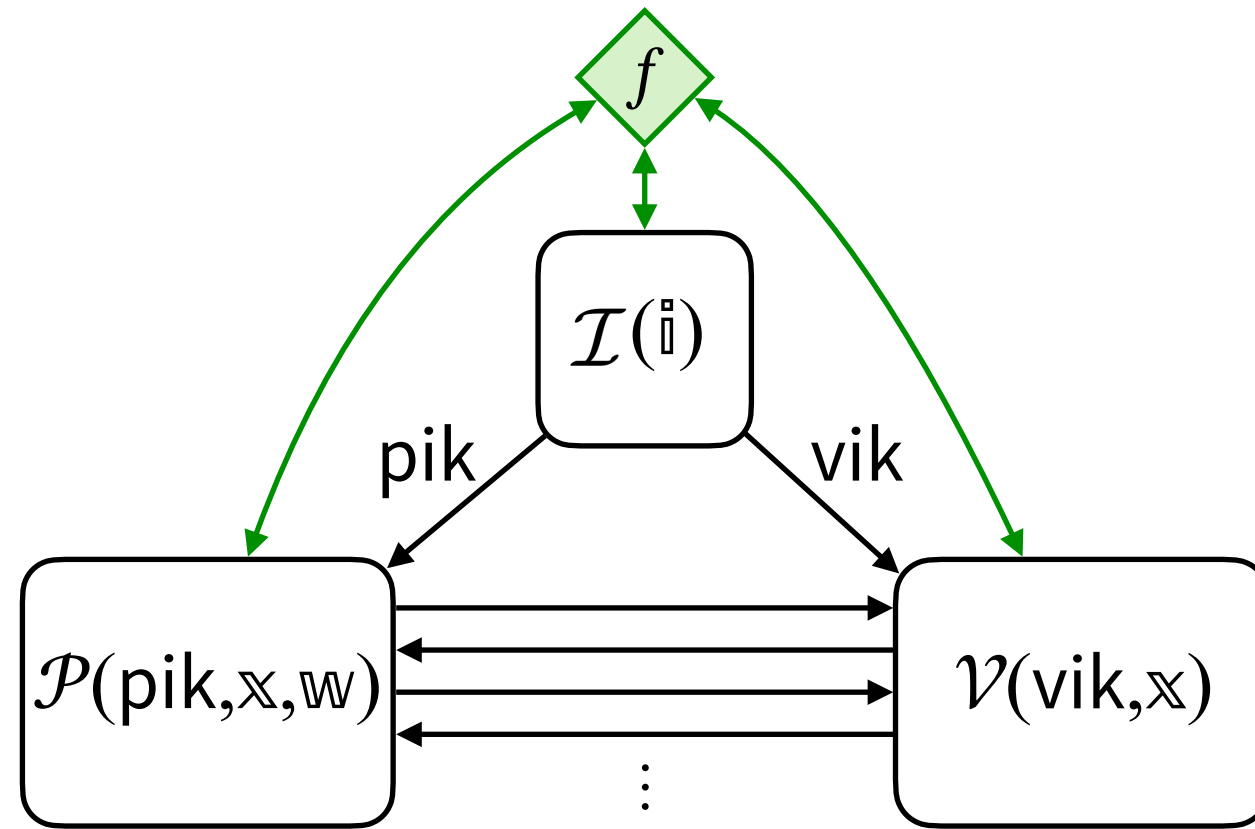


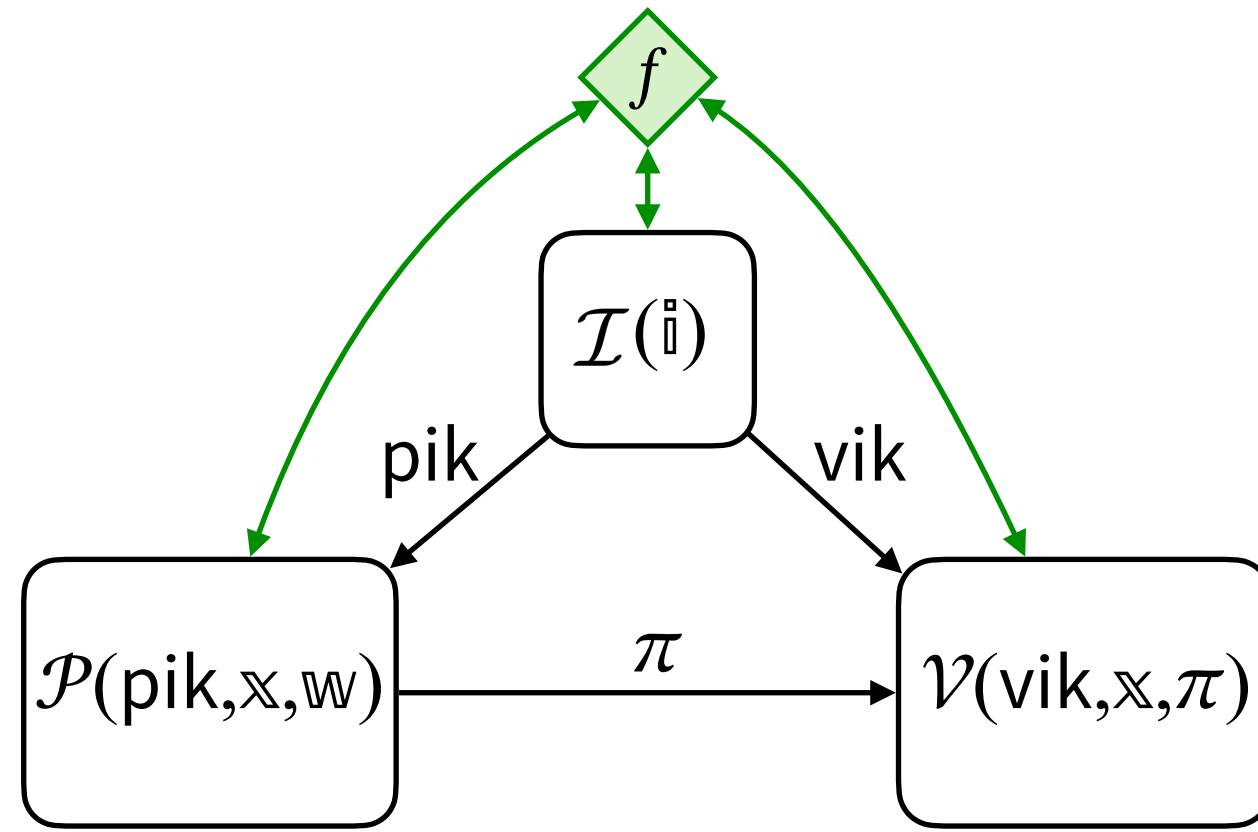


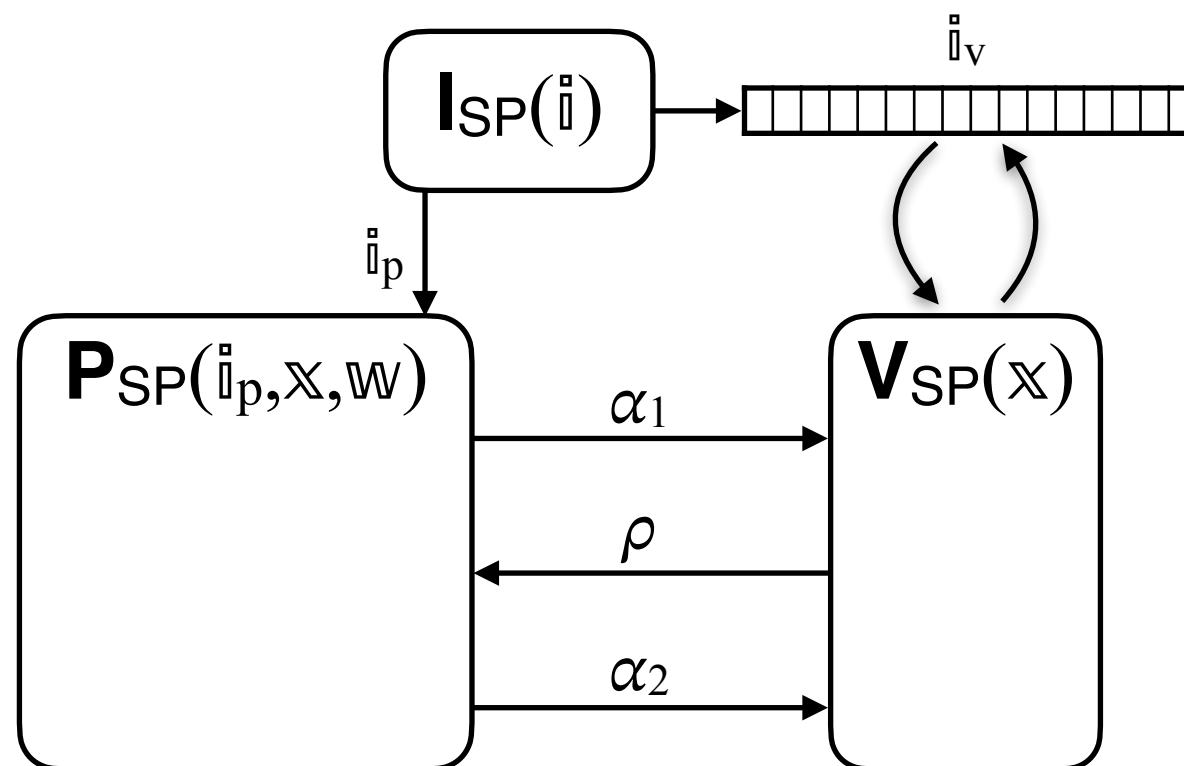


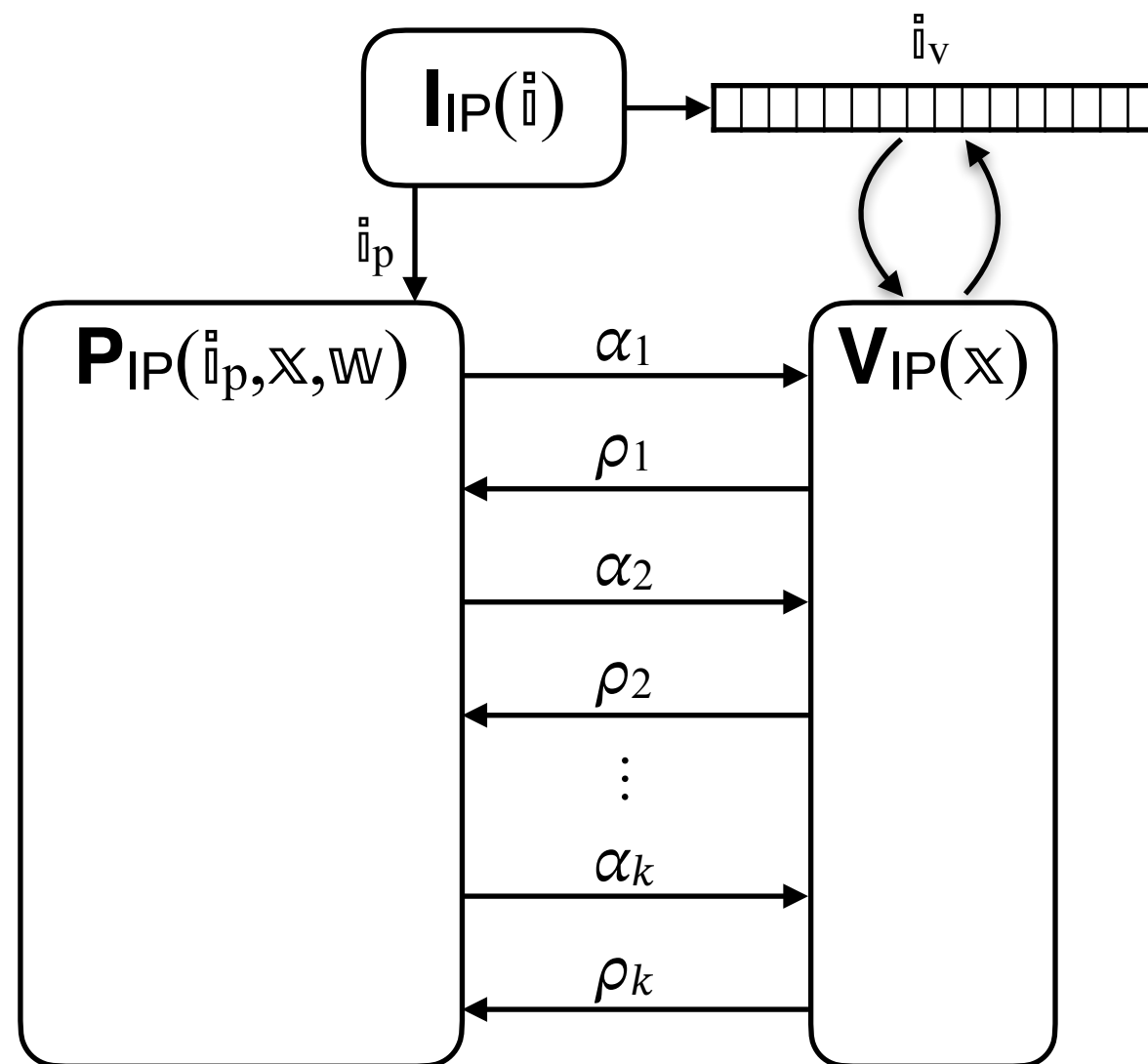


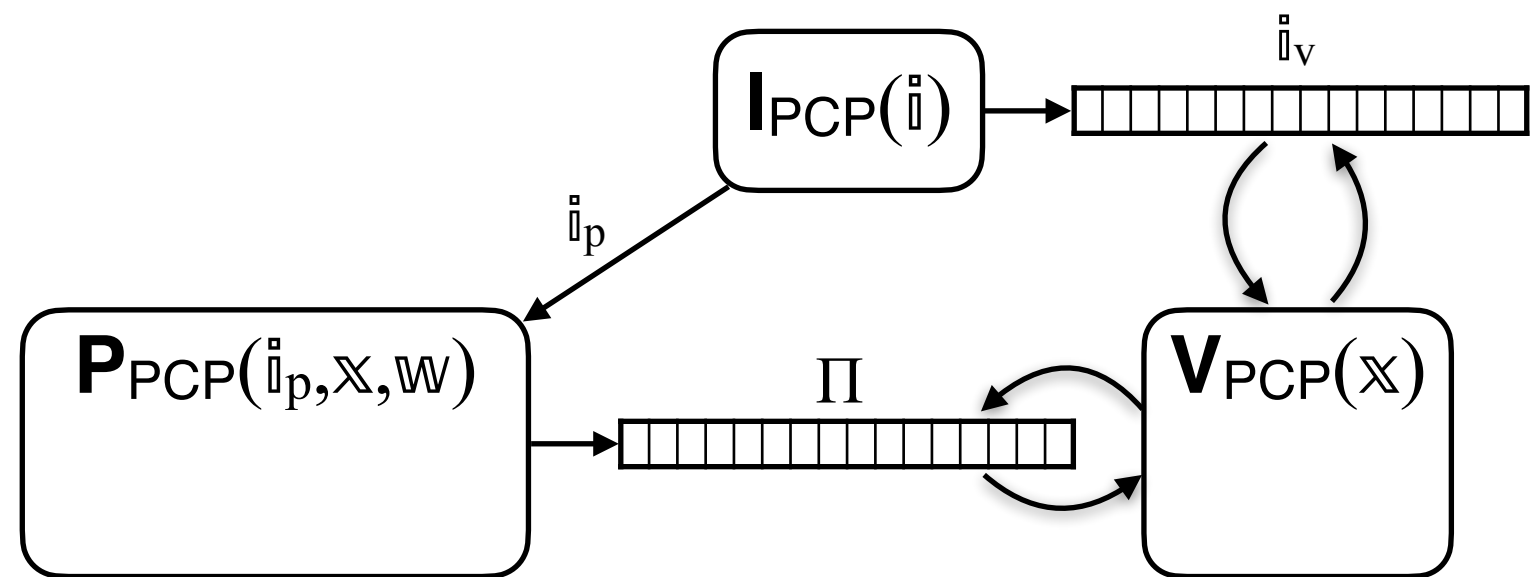


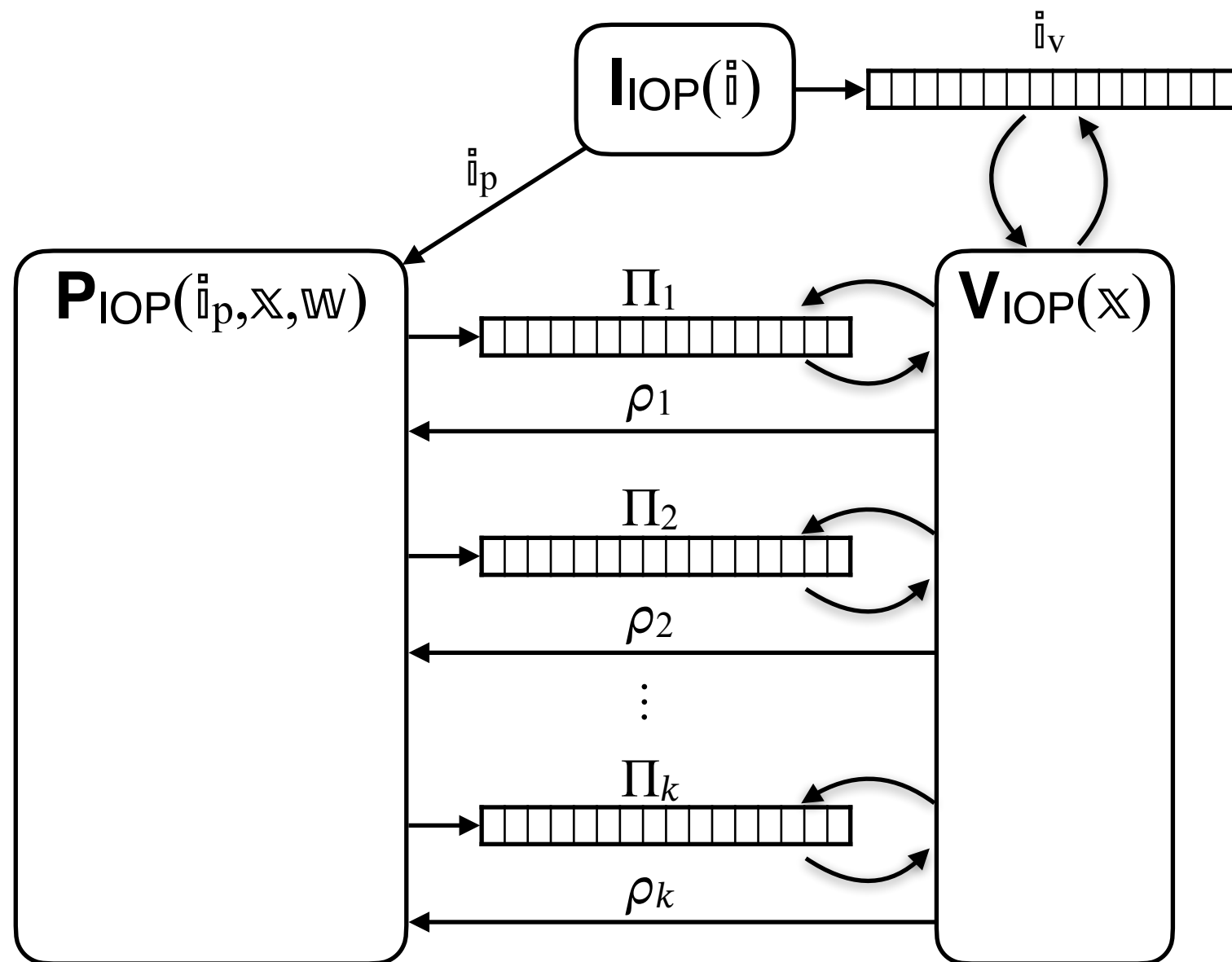




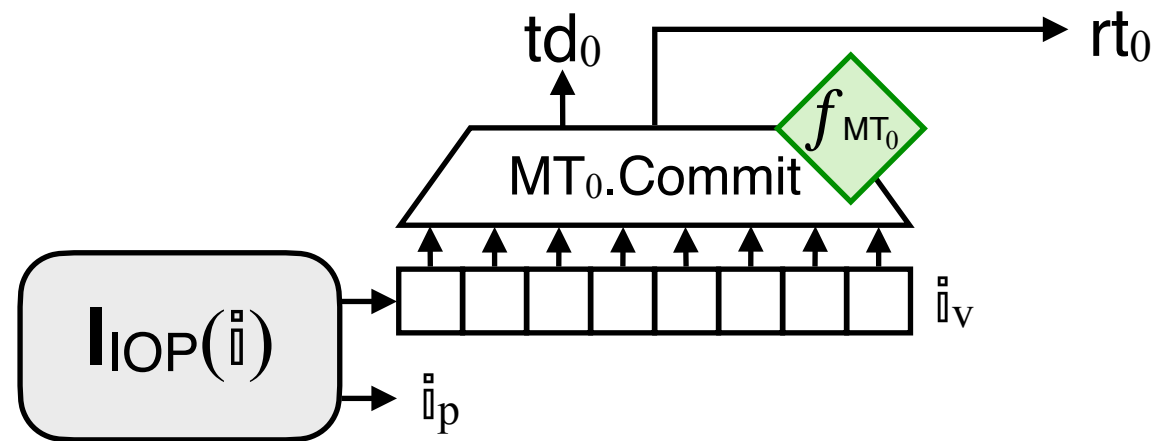
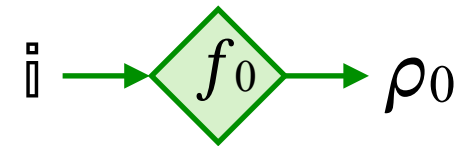








$\mathcal{I}(\mathring{\mathbf{i}})$



$\text{pik} := (\mathring{\mathbf{i}}_p, \mathring{\mathbf{i}}_v, \text{rt}_0, \text{td}_0, \rho_0)$

$\text{vik} := (\text{rt}_0, \rho_0)$

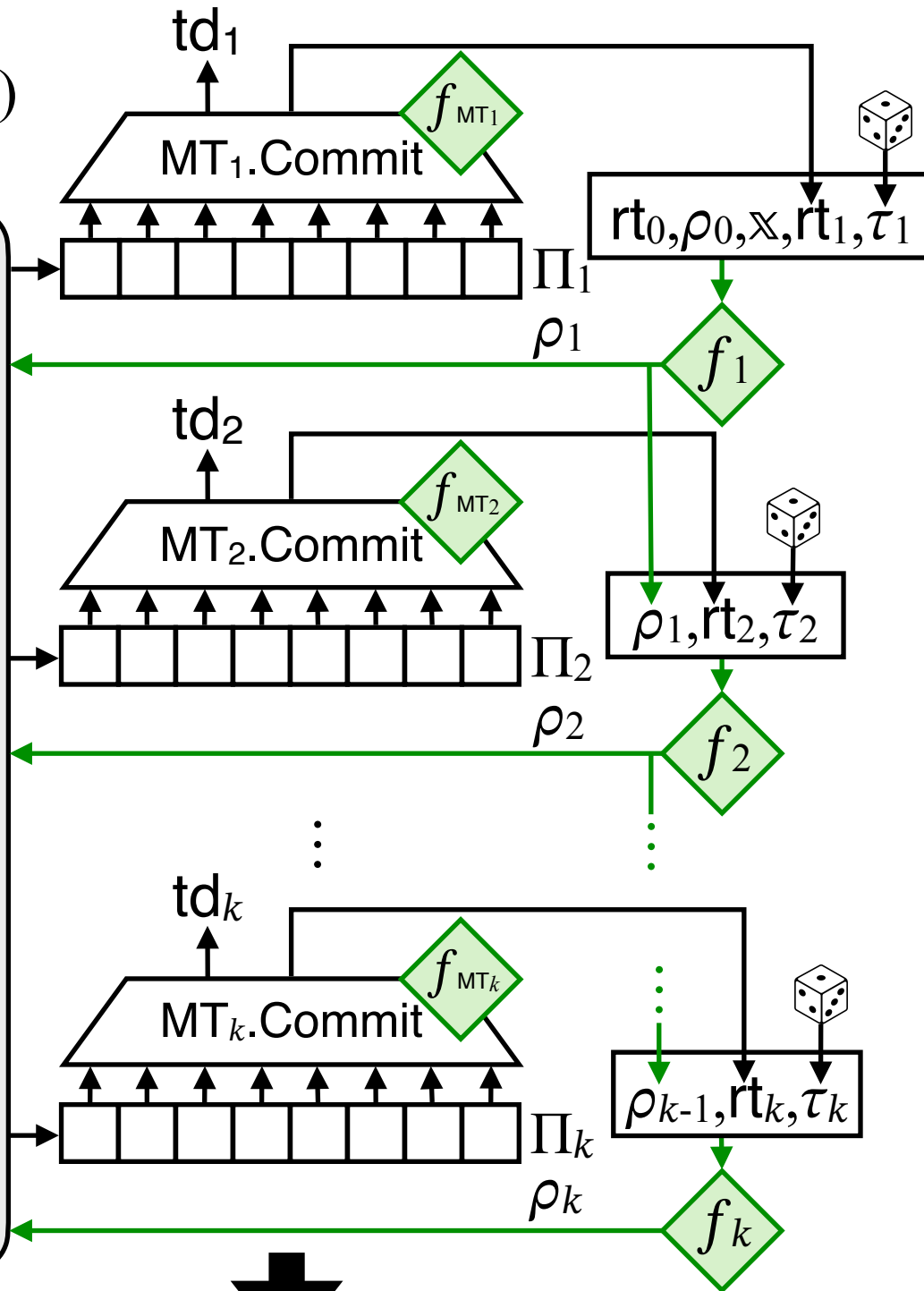
$\downarrow$
pik

$\downarrow$
vik

$\mathcal{P}(\text{pik}, \mathbb{X}, \mathbb{W})$

- parse pik as $(\mathbb{I}_p, \mathbb{I}_v, \text{rt}_0, \text{td}_0, \rho_0)$

$\mathbf{P}_{\text{IOP}}(\mathbb{I}_p, \mathbb{X}, \mathbb{W})$



IOP verifier queries: $(Q_0, Q_1, \dots, Q_k)$

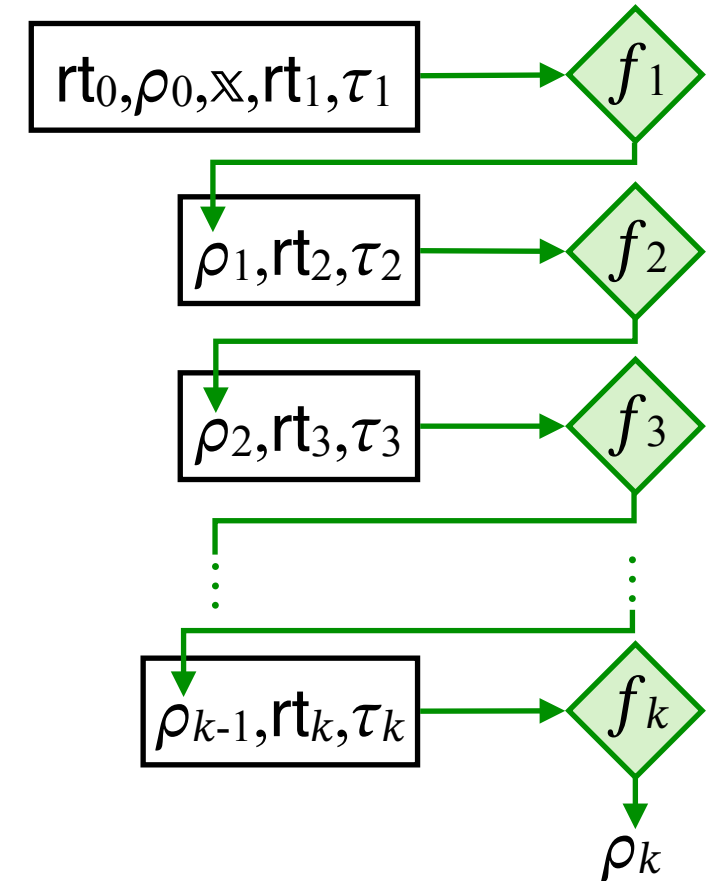
IOP oracle answers: $(\mathbf{a}_0, \mathbf{a}_1, \dots, \mathbf{a}_k)$

MT proofs: $(\text{pf}_0, \text{pf}_1, \dots, \text{pf}_k)$

$\pi := ((Q_0, \mathbf{a}_0, \text{pf}_0), ((\text{rt}_i, Q_i, \mathbf{a}_i, \text{pf}_i, \tau_i))_{i \in [k]})$

$\mathcal{V}(\text{vik}, \mathbb{X}, \pi)$

- parse vik as (rt_0, ρ_0)
- parse π as $((Q_0, \mathbf{a}_0, \text{pf}_0), ((\text{rt}_i, Q_i, \mathbf{a}_i, \text{pf}_i, \tau_i))_{i \in [k]})$
- derive IOP randomness

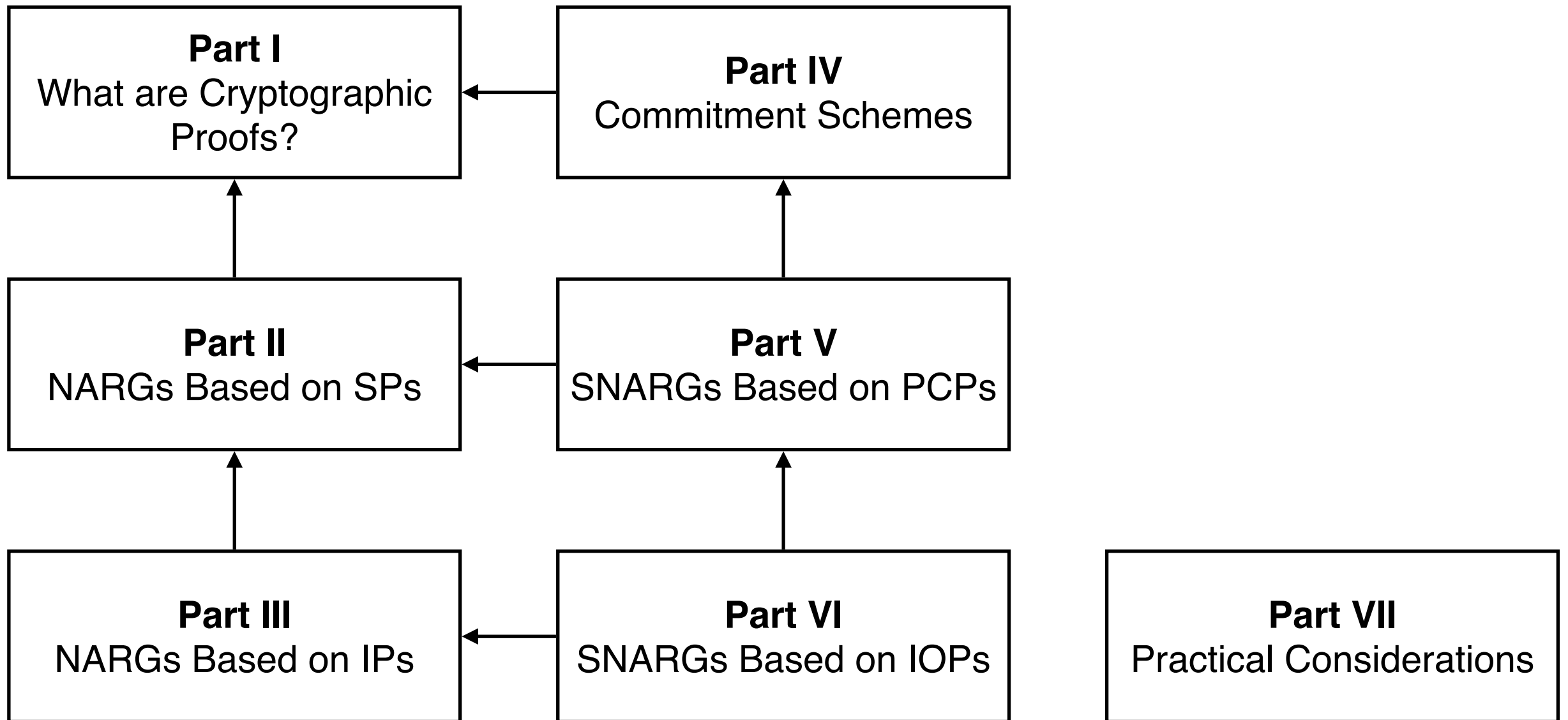


- check MT proofs

$\bigwedge_{i=0}^k \text{MT}_i.\text{Check}(\text{rt}_i, Q_i, \mathbf{a}_i, \text{pf}_i)$

- check IOP decision

$\mathbf{V}_{\text{IOP}}[Q_i, \mathbf{a}_i]_{i=0}^k(\mathbb{X}, (\rho_1, \dots, \rho_k))$



Building Cryptographic Proofs from Hash Functions

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